

Specificity Foundation Models

Antibody-Antigen is All You Need

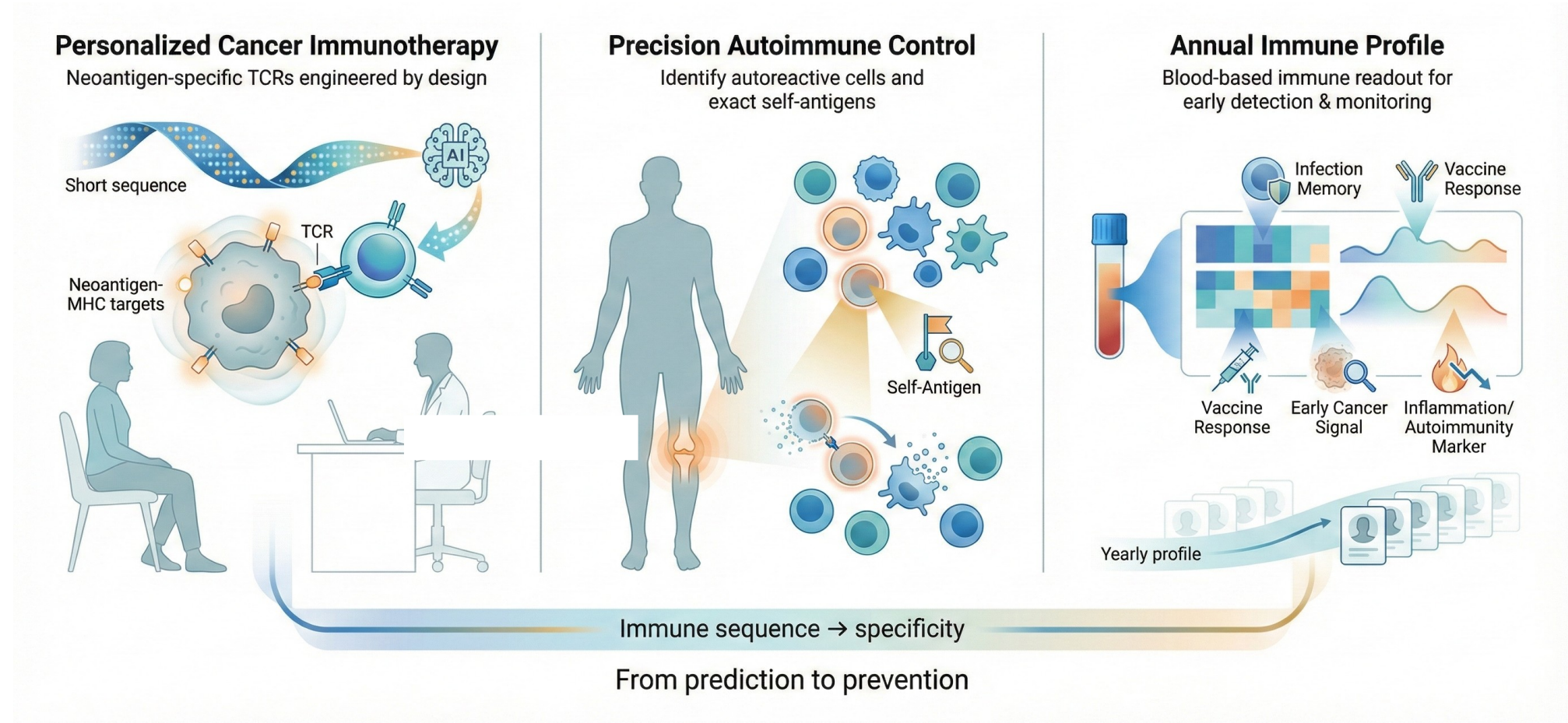
Sai Reddy

Full Professor, ETH Zurich

Scientific Director, Botnar Institute of Immune Engineering (BIIE)

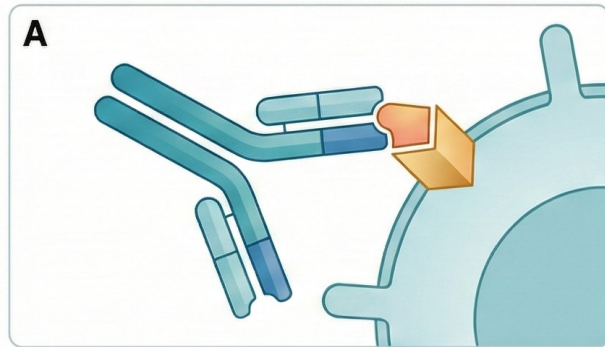
The Potential Future Standard of Care with AI

If AI made it possible to predict Sequence-to-Specificity of Antibodies and TCRs

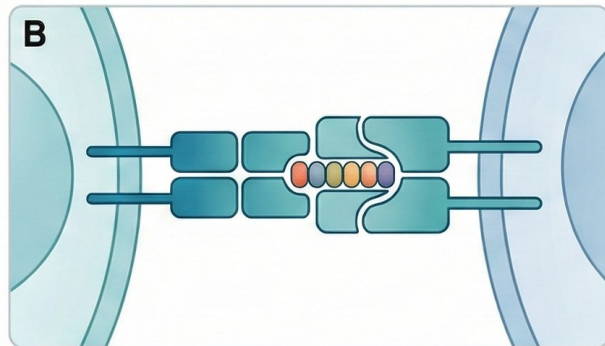


Sequence-to-Specificity

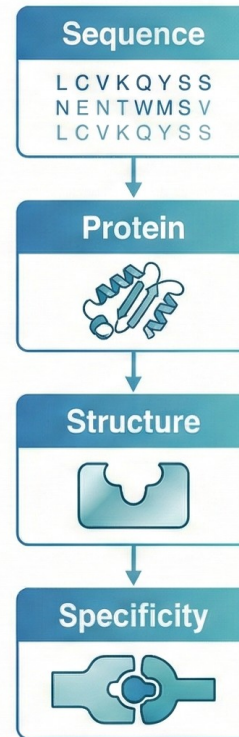
As foundational to Immunology as AlphaFold was to structural biology



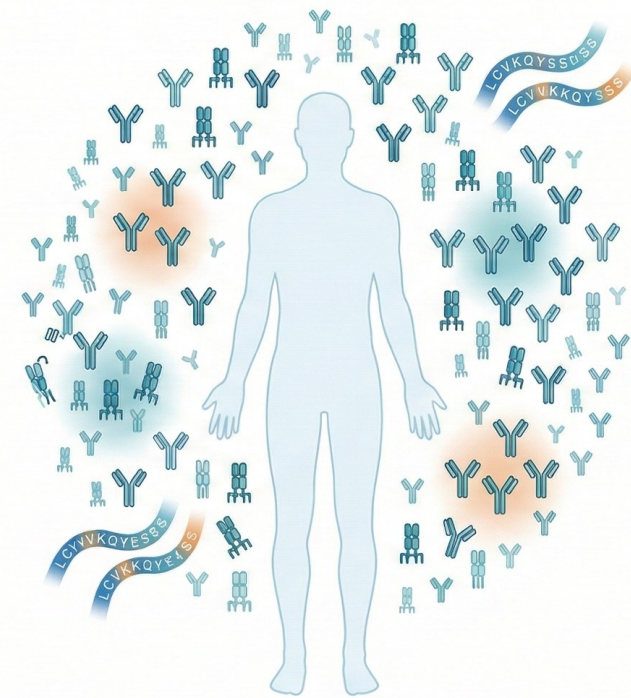
Antibody binds antigen



TCR recognizes peptide-MHC



Sequence determines structure;
structure determines specificity



Immune repertoire
Millions of unique receptors per person

AI is reshaping Drug Discovery

Alphabet's AI biotech Isomorphic Labs bags \$2.1B series B to fuel next-gen drug design model

By James Waldron · May 12, 2026 11:45am

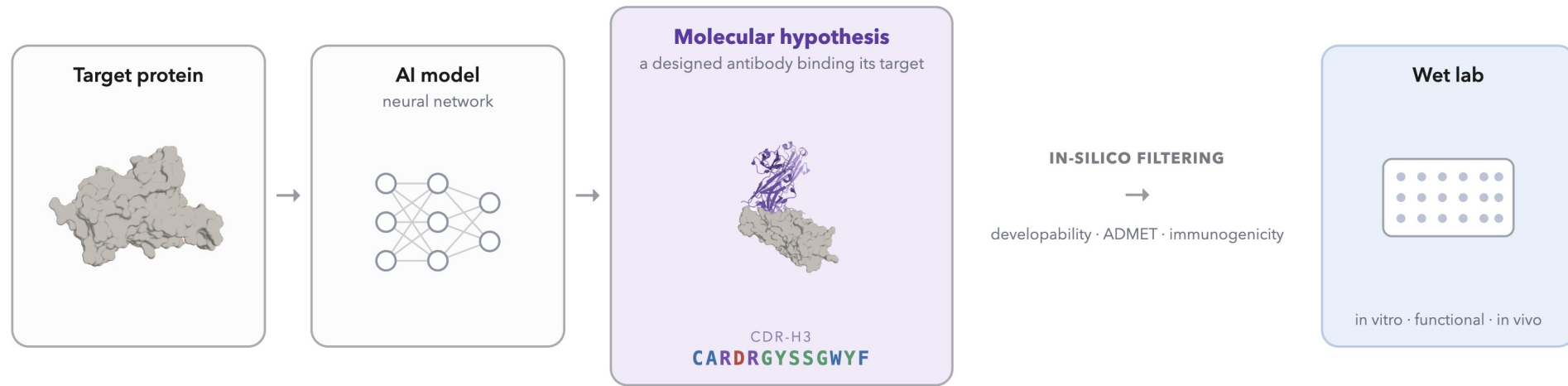
New AI drug discovery powerhouse Xaira rises with \$1B in funding

By Annalee Armstrong · Apr 24, 2024 6:00am

Chai Discovery is working with major drugmakers to use its latest antibody AI model to find new therapies. Now it's in talks to raise \$400 million more VC funding at a \$3.4 billion valuation, Forbes has learned.

The field is betting big on AI drug discovery — but what does it actually deliver?

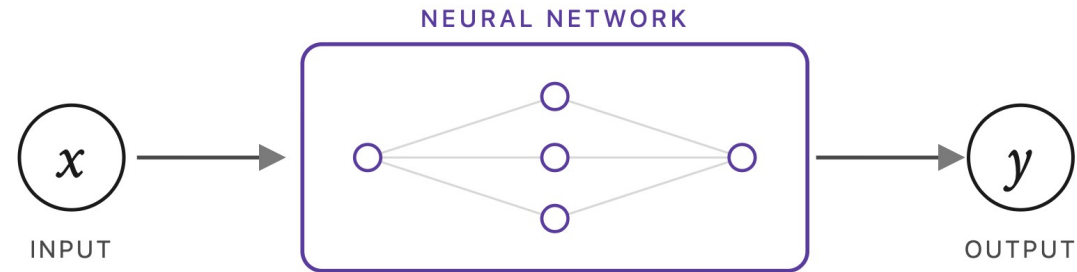
AI Drug Discovery is a Molecular Hypothesis



AI only outputs a molecular hypothesis of binding a target
The experimental wet-lab is where drug development happens

AI is a Fitted Function

A giant equation that is solved for by trial-and-error



$$y = f(x) = w_0 + \sum_i w_i \sigma(a_i x + b_i)$$

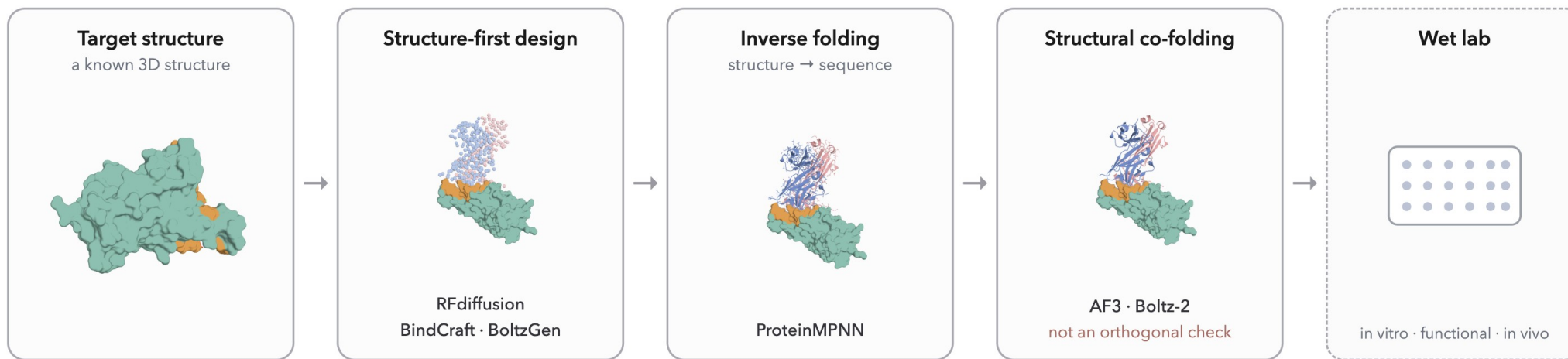
inside the box is just a big equation — one tunable knob (parameter) per connection

UNIVERSAL APPROXIMATION THEOREM

A neural network with enough neurons can approximate **any** continuous function to arbitrary accuracy.

Cybenko 1989 · Hornik, Stinchcombe & White 1989

Structure-based AI Drug Discovery



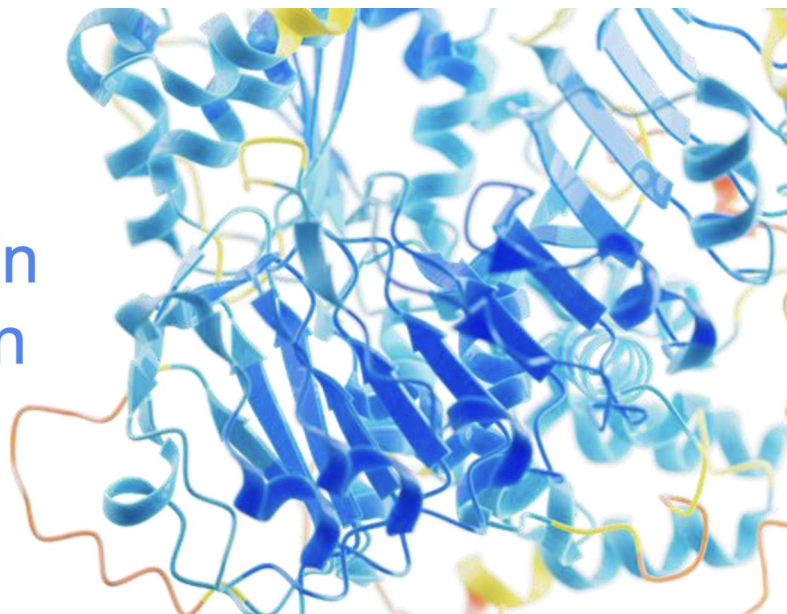
The canonical structure-first approach builds a backbone, designs a sequence for it, then validates by re-predicting structure — powerful and fast-improving

AI models, however, are a **black box**, not interpretable directly

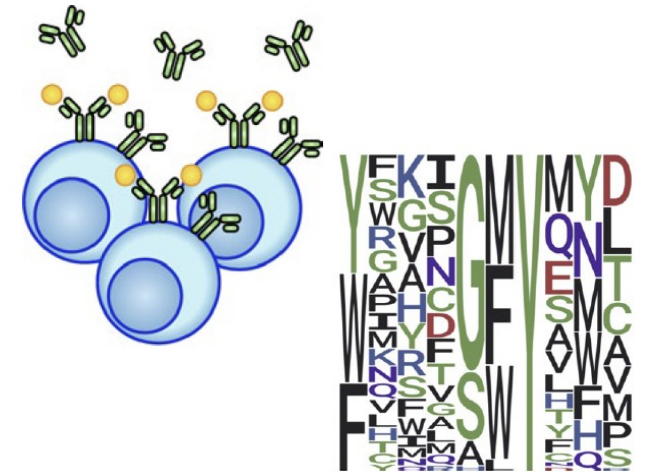
Building chemical and biological intuition into protein structure prediction

John Jumper

Google DeepMind

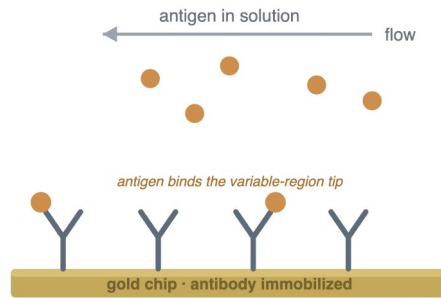


What if we could build immunological intuition
into antibody-antigen specificity prediction?



Antibody-Antigen Binding is determined by Thermodynamics

Statistical mechanics of molecular recognition

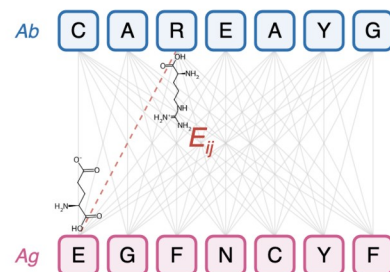


$$K_d = [Ab] [Ag] / [Ab:Ag] = k_{off} / k_{on}$$

$$\Delta G^\circ = RT \ln K_d$$

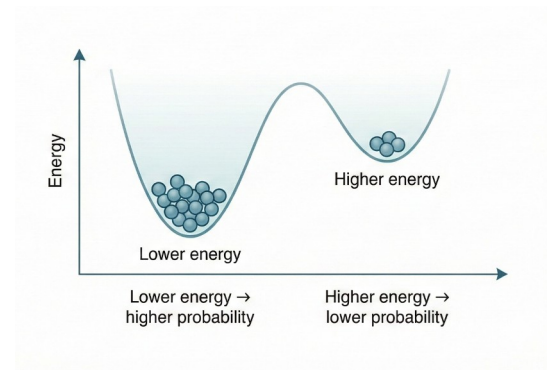
$$\Delta G = \Delta H - T\Delta S$$

$$\Delta G = E_{ij}$$

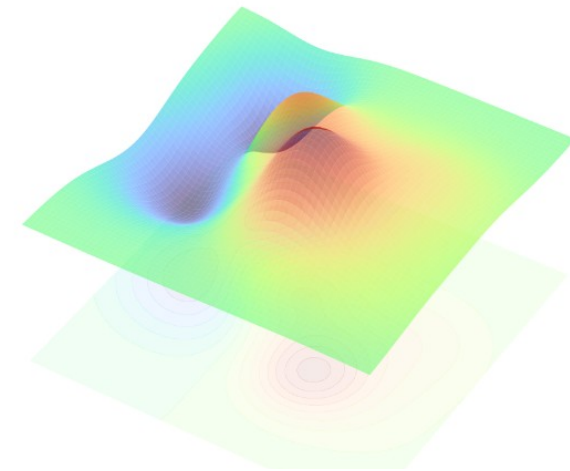


Ludwig Boltzmann, 1877

$$P(j | i) = \frac{e^{-E_{ij}/k_B T}}{\sum_{j'} e^{-E_{ij'}/k_B T}}$$



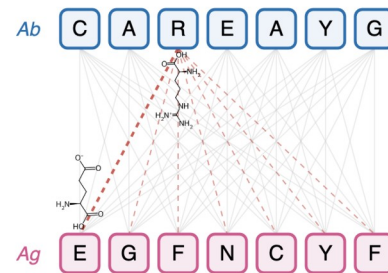
$$E_{ij} = \Delta G$$



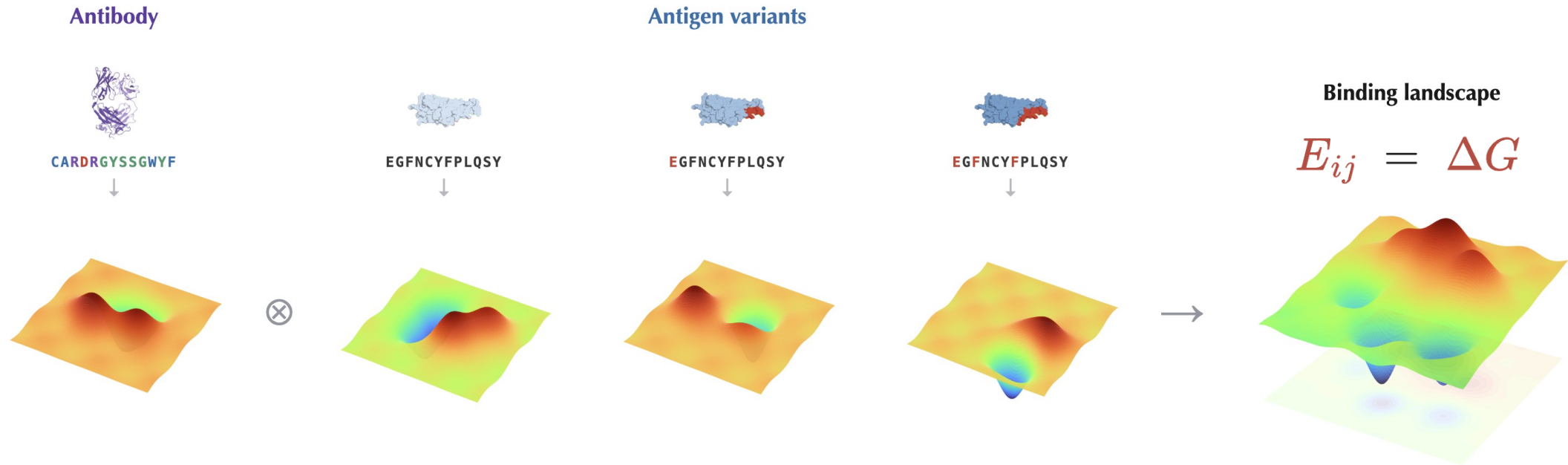
Second Law of Thermodynamics Boltzmann Distribution

At equilibrium, a systems must move towards lowest energy states.

Not a choice — a law.



Every Molecule has its own Energy Landscape



An energy landscape isn't a property of a pair – each molecule carries one

Every molecule (Ab or Ag) → its own, high-dimensional landscape

Landscape represents a collection of biophysical and thermodynamic properties

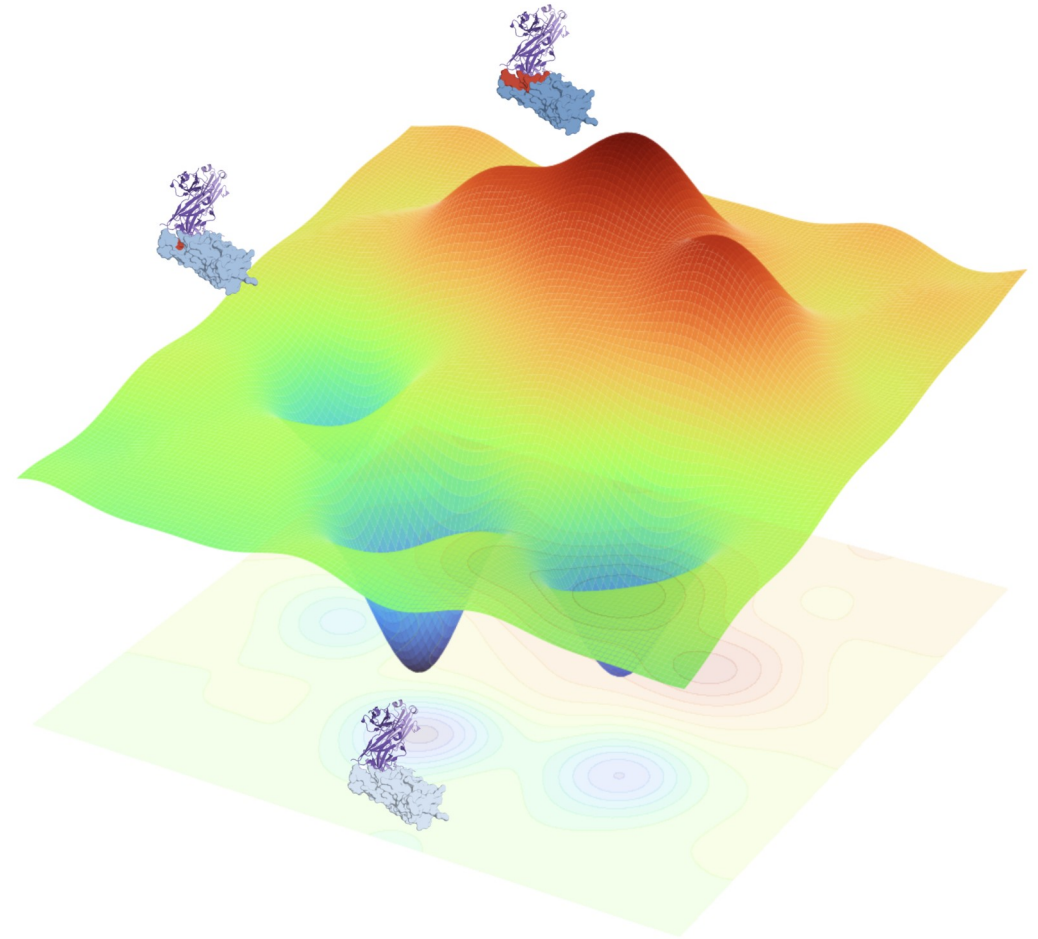
Binding landscape = antibody landscape $\otimes$ antigen landscape

The Binding Free Energy Landscape

1. How can we represent the Energy Landscape?
2. How can we calculate a Binding Energy Landscape?

$$P(j \mid i) = \frac{\exp(-E_{ij}/k_B T)}{\sum_{j'} \exp(-E_{ij'}/k_B T)}$$

$$E_{ij} = \Delta G = \Delta H - T \cdot \Delta S$$



Same antibody, three antigens, each with different energy
Deep funnel = low ΔG = tight (cognate)
Ridge = high ΔG = weak (escape)

Large Language Models determine an Energy Landscape

SENTENCE • LLM

all possible sentences (astronomically many)

the cat sat on the manifold
why blue of the running not

the cat sat on the **mat** **EXISTS**

ideas sleep furiously green

the web keeps the ones
selected by **meaning & usage**

PROTEIN • PLM

all possible sequences (20^L)

WQPVMLKKDTRH

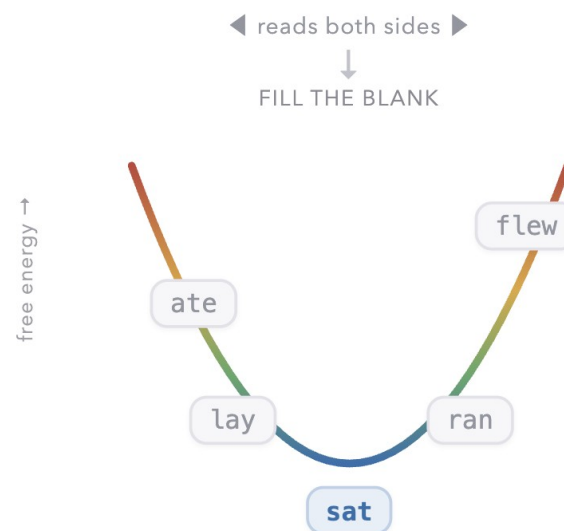
GGRRPPLLDKMQA

CARDRGYSSGWYF **EXISTS**

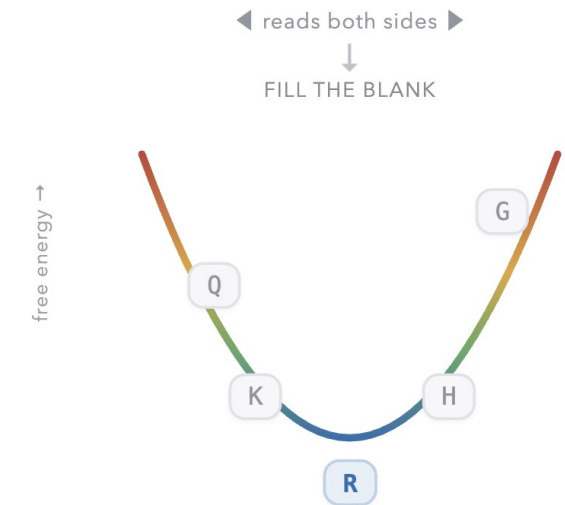
ZKMNQVQWXLPEY

evolution keeps the ones
selected by **thermodynamics** (folding ΔG)

SENTENCE • LLM



ANTIBODY CDR-H3 • PLM



The model's log-probabilities are the energies:
it learned them by predicting nature

$$\mathcal{L} = -\log P(\text{aa} \mid \text{context})$$

$$E = -k_B T \log P$$

PLMs represent a High-dimensional Energy Landscape

one amino acid becomes a **high-dimensional vector** — a list of **1280** numbers



Molecular Binding is Computed by an Energy Dot Product

Free energy can be measured for DNA, for antibodies and proteins it is learned

DNA-DNA: every base energy can be measured

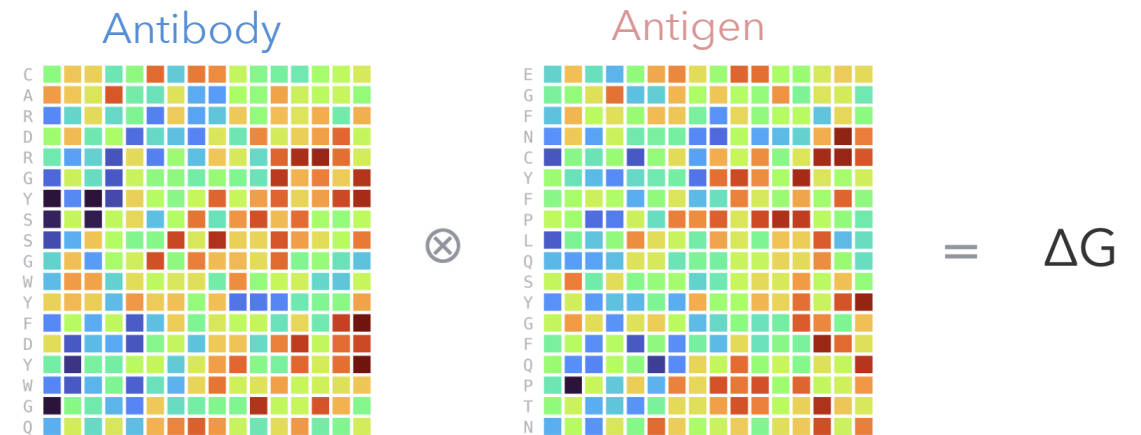
ΔG is the dot product in thermodynamic space



	GC	CT	TT	TT	TA	COUNT
GC / CG	-2.24					1
CT / GA		-1.28				1
AA / TT			-1.00	-1.00		2
TA / TA					-0.58	1
step →	1	2	3	4	5	

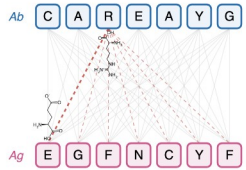
Ab-Ag: energy per residue must be learned

ΔG must be the dot product, same Second Law



$$\Delta G^\circ = 1(-2.24) + 1(-1.28) + 2(-1.00) + 1(-0.58) = -6.10 \text{ kcal/mol}$$

Computing Molecular Binding from the Boltzmann Distribution

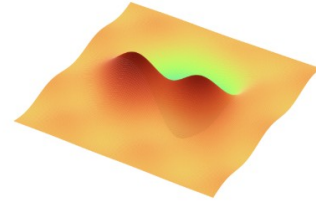


$$P(j | i) = \frac{\exp(-E_{ij}/k_B T)}{\sum_{j'} \exp(-E_{ij'}/k_B T)}$$

$$P(j|i) = \frac{\exp(-\text{Antibody}_i \cdot \text{Antigen}_j / k_B T)}{\sum_{j'} \exp(-\text{Antibody}_i \cdot \text{Antigen}_{j'} / k_B T) + \exp(-\text{Antibody}_i \cdot \text{Antigen}_j / k_B T) + \exp(-\text{Antibody}_i \cdot \text{Antigen}_{j''} / k_B T) + \dots}$$

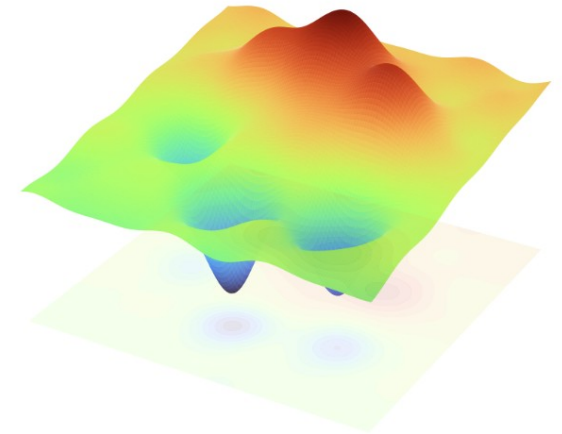
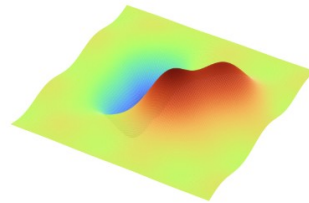
Antibody

C A R D R G Y



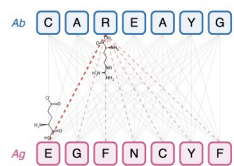
Antigen

E G F N C Y F

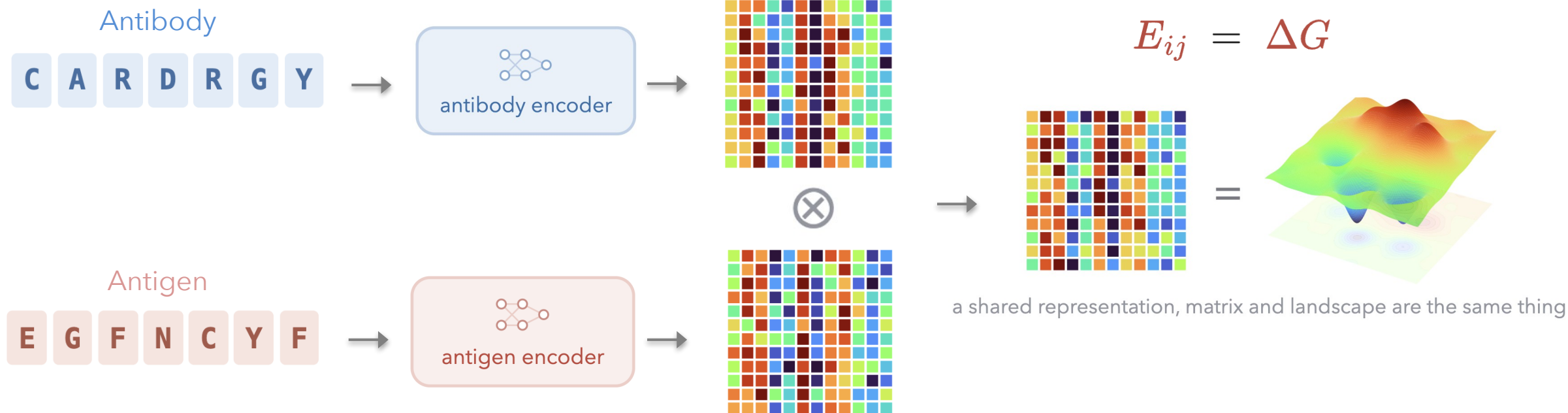


$$E_{ij} = \Delta G$$

Computing Molecular Binding from the Boltzmann Distribution



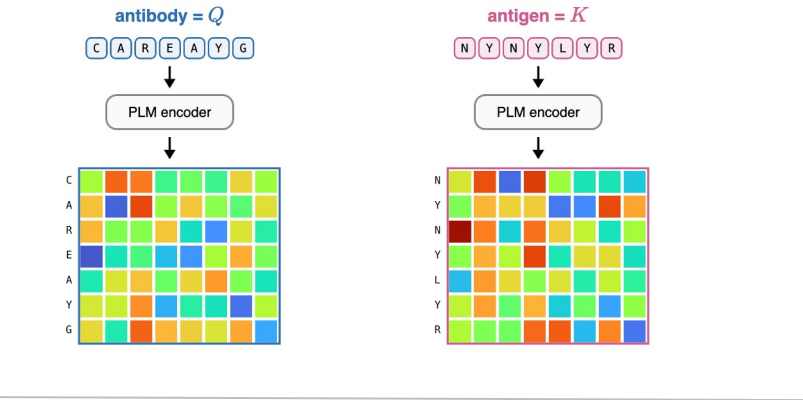
$$P(j | i) = \frac{\exp(-E_{ij}/k_B T)}{\sum_{j'} \exp(-E_{ij'}/k_B T)}$$



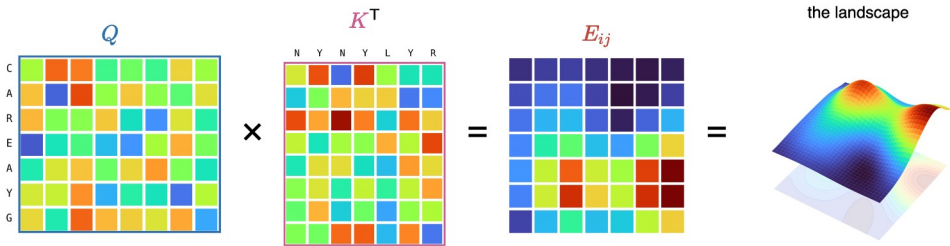
Boltzmann Distribution is Softmax Attention

$$P(j | i) = \frac{e^{-E_{ij}/k_B T}}{\sum_{j'} e^{-E_{ij'}/k_B T}}$$

L. Boltzmann, 1877



the dot product of the matrices is the binding landscape



every entry is a single dot product $E_{ij} = \mathbf{q}_i \cdot \mathbf{k}_j$ — antibody residue i · antigen residue j

substitute into Boltzmann $\cdot k_B T = \sqrt{d}$

$$P(j | i) = \frac{e^{\mathbf{q}_i \cdot \mathbf{k}_j / \sqrt{d}}}{\sum_{j'} e^{\mathbf{q}_i \cdot \mathbf{k}_{j'} / \sqrt{d}}}$$

the normalized exponential $e^z / \sum_{j'} e^z$ is the softmax

$$P(j | i) = \text{softmax} \left(\frac{Q K^T}{\sqrt{d}} \right)$$

A. Vaswani, 2017

Attention Is All You Need

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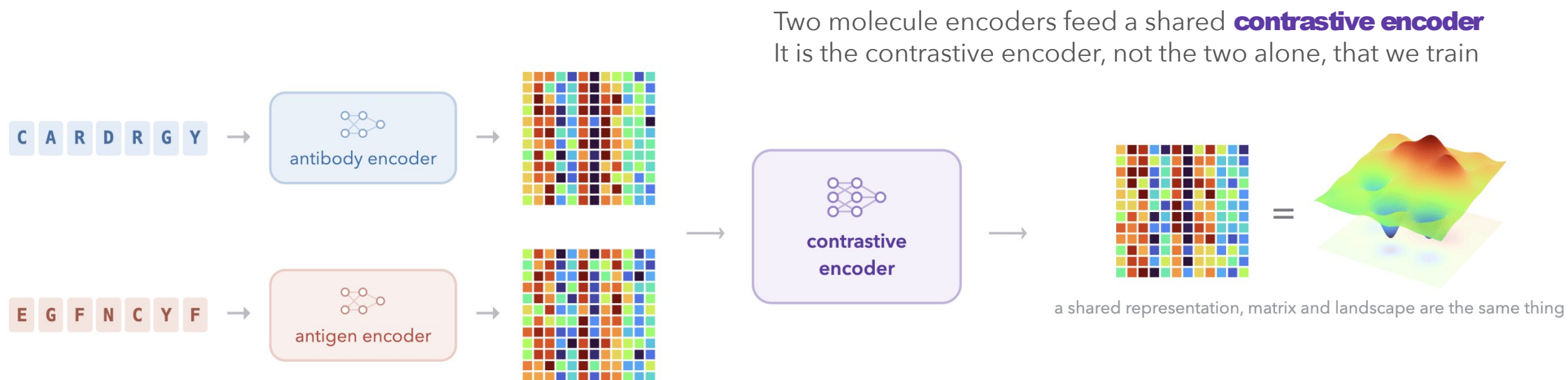
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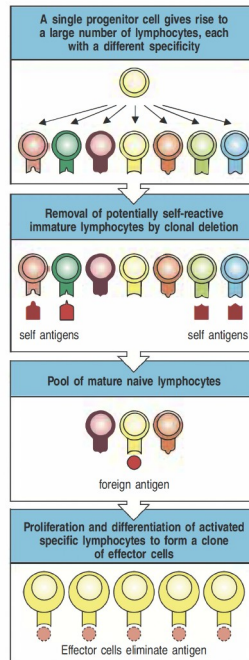
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255,203 citations since 2017

Contrastive Encoder Predicts Antibody-Antigen Binding



Clonal Selection is determined by Thermodynamics

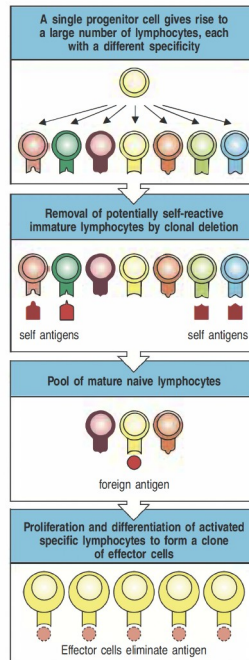


$$P(\text{select}) = \frac{\exp(a(\text{Ab}, \text{Ag})/\tau)}{\exp(a(\text{Ab}, \text{Ag})/\tau) + \sum_i \exp(a(\text{Ab}, \text{Ag}_i)/\tau)}$$

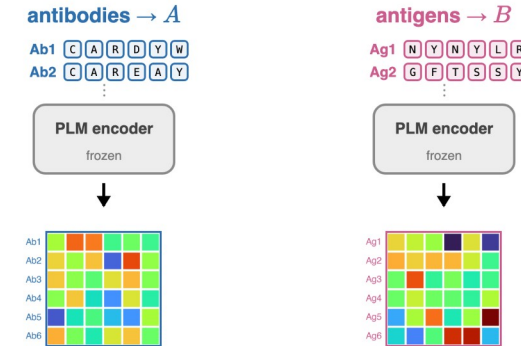
a = binding affinity (antibody to antigen) · τ = selection stringency (temperature)

G. Bell, A. Perelson, 1977

Clonal Selection is determined by Thermodynamics



a frozen PLM encoder embeds many antibodies and antigens into a shared co-embedding space

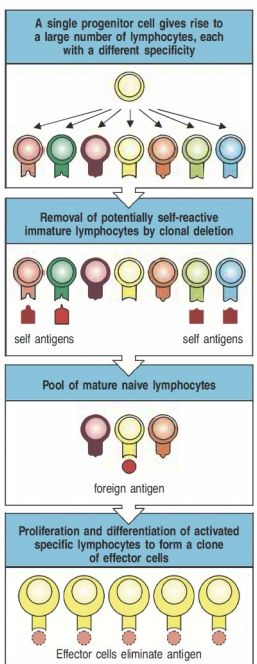


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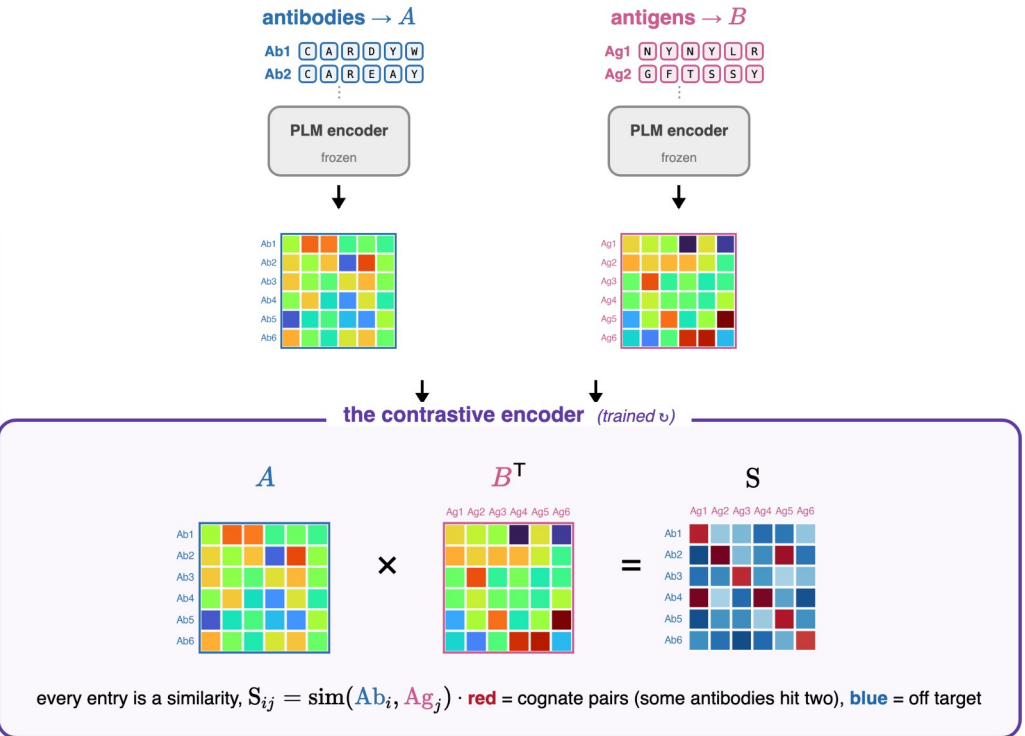
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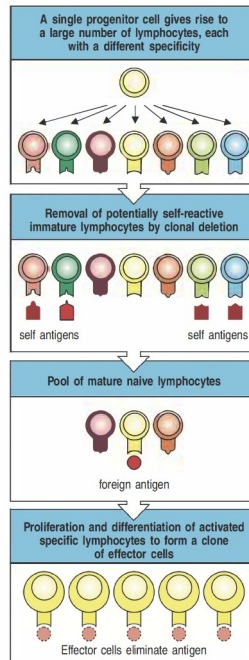
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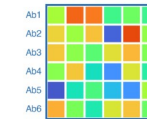
a frozen PLM encoder embeds many antibodies and antigens into a shared co-embedding space

V backprop · trains the encoder

antibodies $\rightarrow A$

Ab1 C A R D Y W
Ab2 C A R E A Y
...

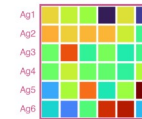
PLM encoder
frozen



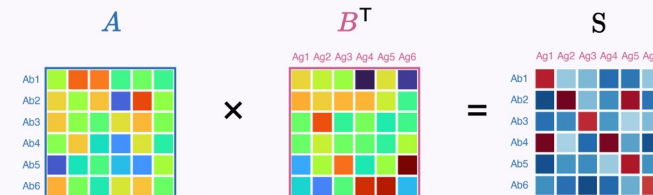
antigens $\rightarrow B$

Ag1 N Y N Y L R
Ag2 G F T S S Y
...

PLM encoder
frozen



the contrastive encoder (trained v)



every entry is a similarity, $S_{ij} = \text{sim}(\text{Ab}_i, \text{Ag}_j)$ · red = cognate pairs (some antibodies hit two), blue = off target

softmax each antibody's row of S, then minimize the negative log of its cognate antigen. that is the InfoNCE loss

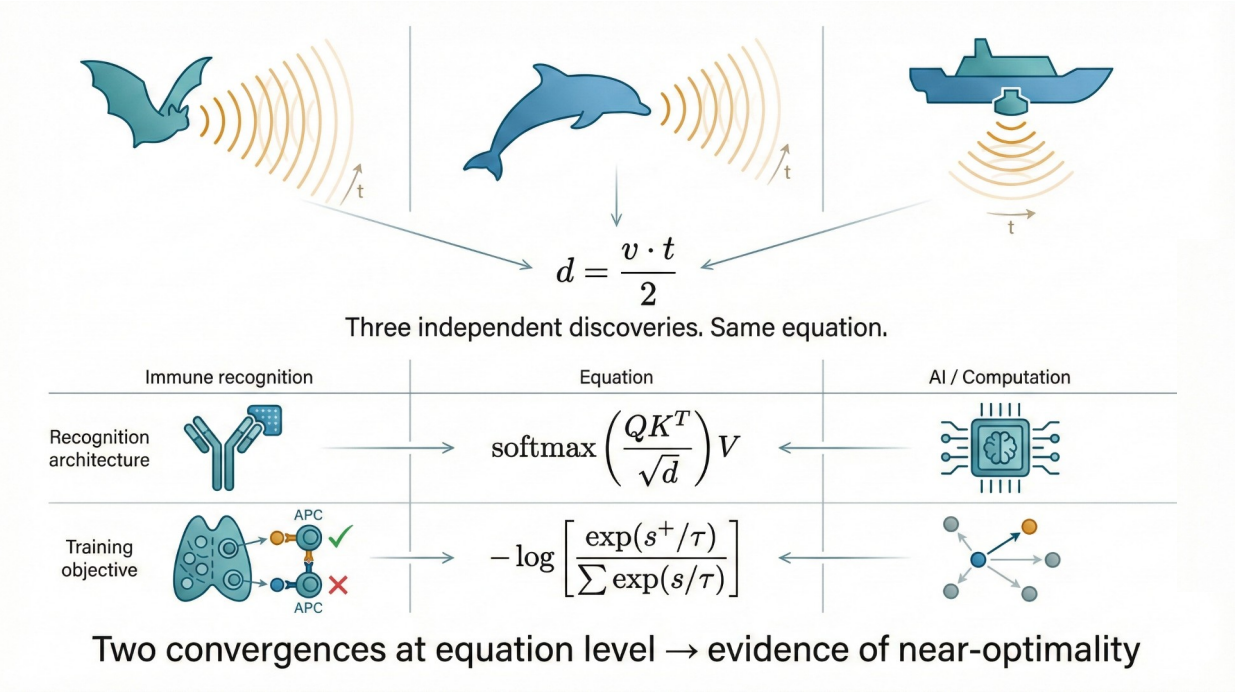
$$\mathcal{L}_{\text{InfoNCE}} = -\log \left[\frac{\exp(\text{sim}(\text{Ab}, \text{Ag})/\tau)}{\exp(\text{sim}(\text{Ab}, \text{Ag})/\tau) + \sum_i \exp(\text{sim}(\text{Ab}, \text{Ag}_i)/\tau)} \right]$$

$$\mathcal{L}_{\text{InfoNCE}} = -\log P(\text{select})$$

A. van den Oord, 2018

Isomorphism of Adaptive Immunity and AI

How 500 million years of evolution solved deep learning



Computational Convergence of Adaptive Immunity and Artificial Intelligence

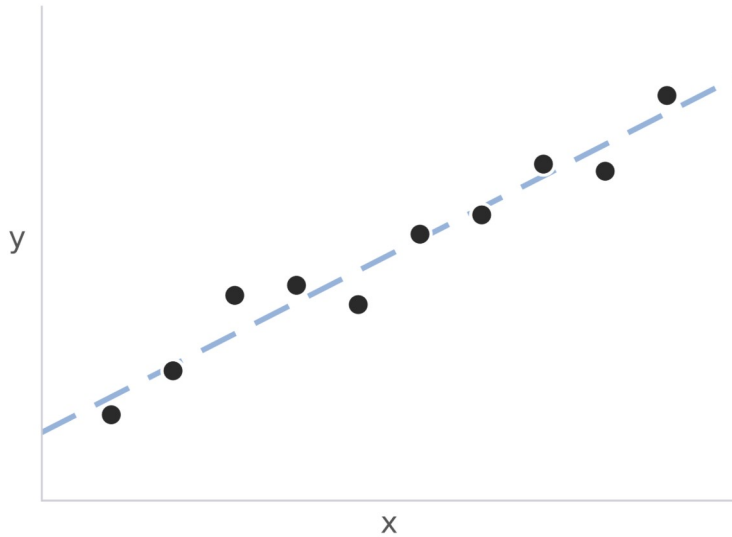
Sai T. Reddy

Posted February 05, 2026.

bioRxiv

AI Architecture	Immune Equivalent	Type
Transformer attention	Boltzmann binding	Exact equivalence
Contrastive learning	Clonal selection	Exact equivalence
Pre-train / Fine-tune / RLHF	Germline / SHM / T _{fh}	Strategic
RAG	Plasma + Memory B cells	Strategic

Knowing the Equation: Parameter Estimation Not Function Approximation



$$y = m x + b$$

A KNOWN EQUATION, TWO PARAMETERS

- 1 Ten points are enough to **solve for m and b**
- 2 The fitted line now **predicts any y from x**, or x from y

An SFM works the same way – the convergence gave the equation, the data fixes its parameters.

Known form → a handful of points pin it

Extreme data efficiency compared to other AI

$$P(j | i) = \frac{e^{-E_{ij}/k_B T}}{\sum_{j'} e^{-E_{ij'}/k_B T}}$$

L. Boltzmann, 1877

$$P(j | i) = \text{softmax}\left(\frac{Q K^T}{\sqrt{d}}\right)$$

A. Vaswani, 2017

Contrastive Learning for Antibody-Antigen Sequence-to-Specificity Prediction

Antibody-Antigen Sequence-to-Specificity

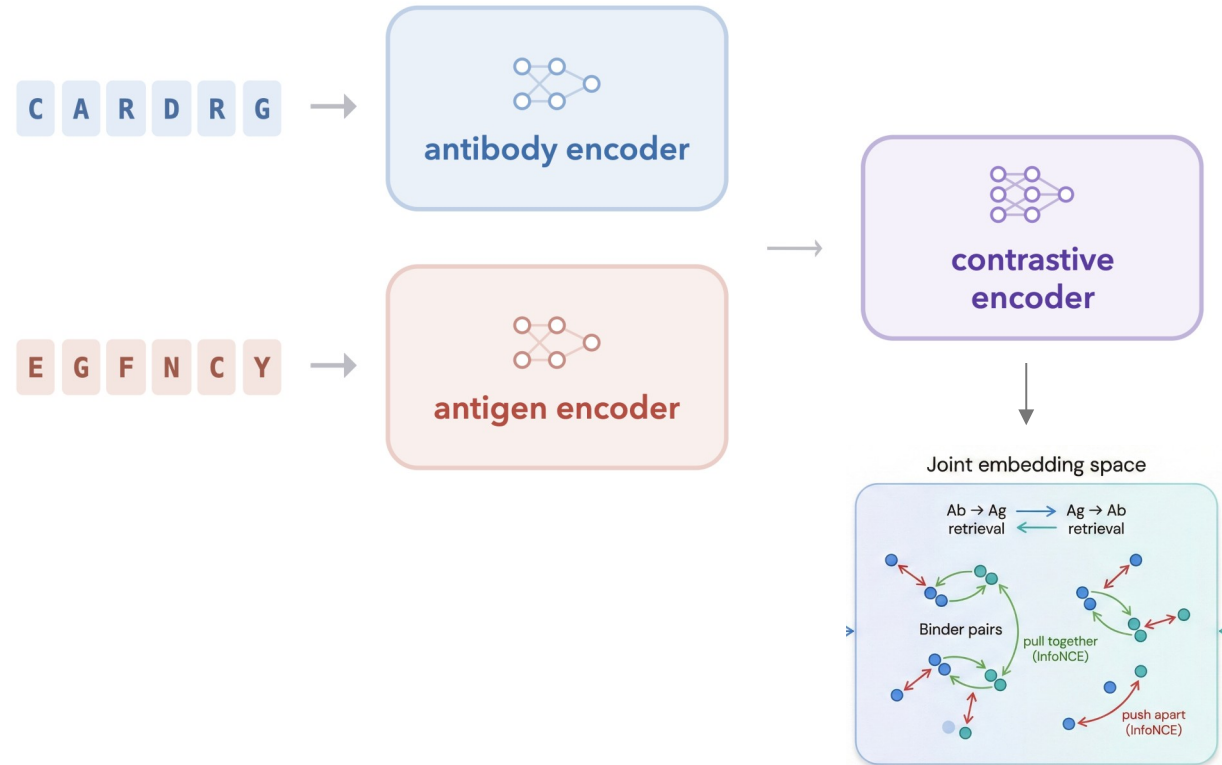
CALM is first prototype model

Antibody Encoder (Antiberty)

Antigen Encoder (ESM2)

Contrastive Encoder with InfoNCE

Training data is 4,139 Ab-Ag pairs (SAbDab)

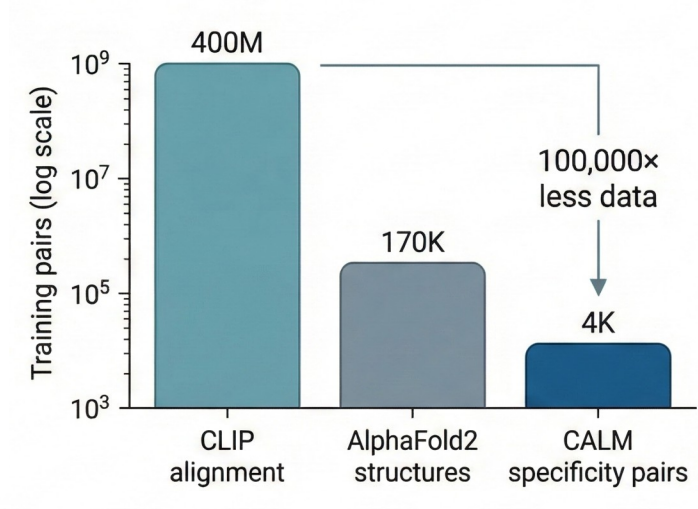
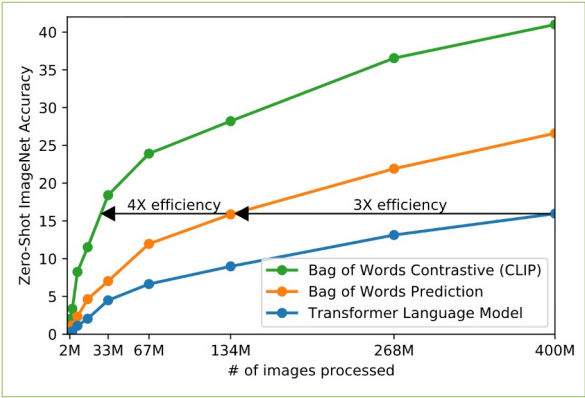
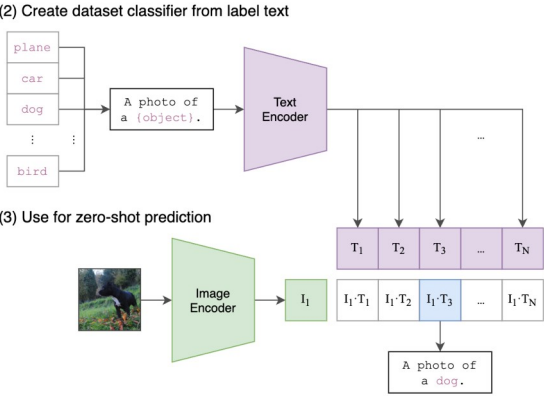


A Closer Look at Contrastive Learning

Data requirements and previous work

Learning Transferable Visual Models From Natural Language Supervision

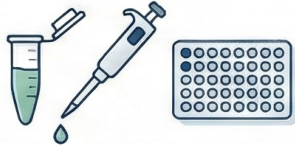
Alec Radford^{*1} Jong Wook Kim^{*1} Chris Hallacy¹ Aditya Ramesh¹ Gabriel Goh¹ Sandhini Agarwal¹
Girish Sastry¹ Amanda Askell¹ Pamela Mishkin¹ Jack Clark¹ Gretchen Krueger¹ Ilya Sutskever¹



Model	Task	Mapping	Data
CLIP	Vision-language alignment	Many-to-one	400M
AlphaFold2	Structure prediction	Many-to-one	170K
CALM	Binding specificity	Many-to-one	4K

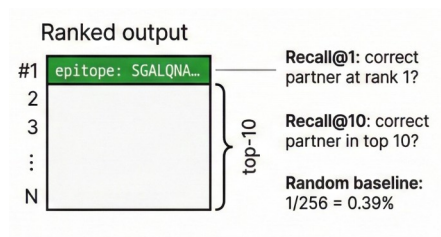
Retrieval: How would Match 256 x 256 Antibody-Antigens

Only sequences provided

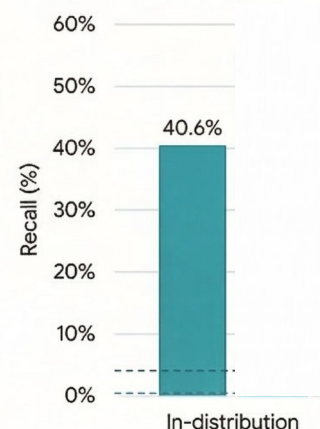
	Experimental (wet lab) 
What it does	Measure binding for every pair
Input required	Physical protein
Operations	65,536 binding measurements
Time (256x256)	Weeks to months
Compute cost	N/A
Reagent cost	\$50K–\$200K
Output	Yes/No per pair
Scales to 1Kx1K?	Months
Scales to 10Kx10K?	Impossible

CALM Performance with only 4,000 Training Pairs

Only sequences provided



SAbDab-only • bidirectional average (Ab→Ag and Ag→Ab)



R@1: correct antigen ranked #1
R@10: correct antigen in top 10

■ R@1

Wet-lab interpretation

R@1 = 40%

For 102 of 256 novel antibodies, CALM identifies the correct antigen on the first guess

R@10 = 55%

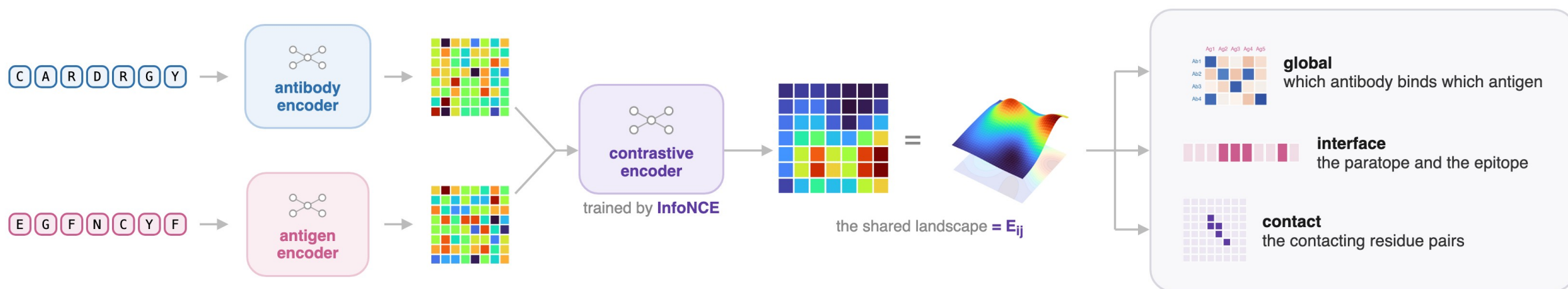
For 141 of 256, correct partner is in the top 10
→ screen 10 candidates, not 256.

Results from: Lee et al, *bioRxiv*, 2026 → [Table 3](#).

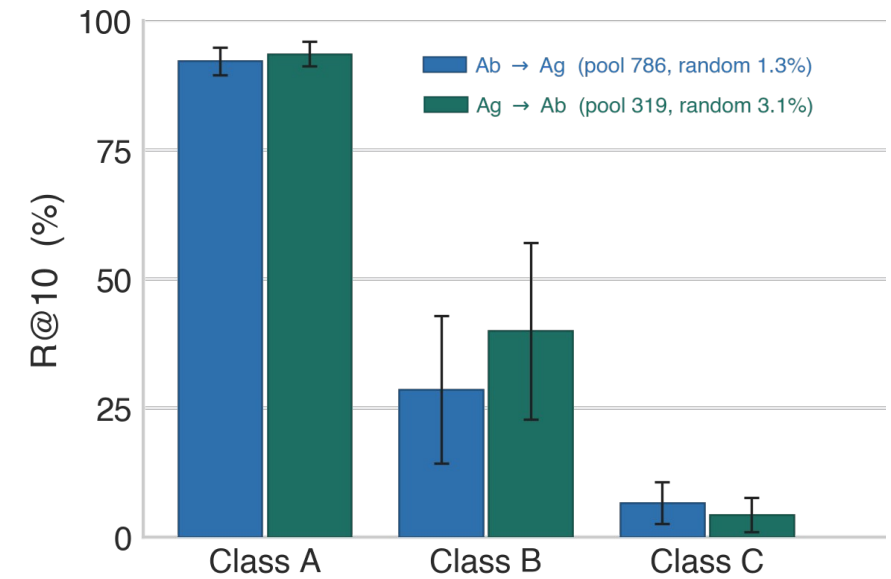
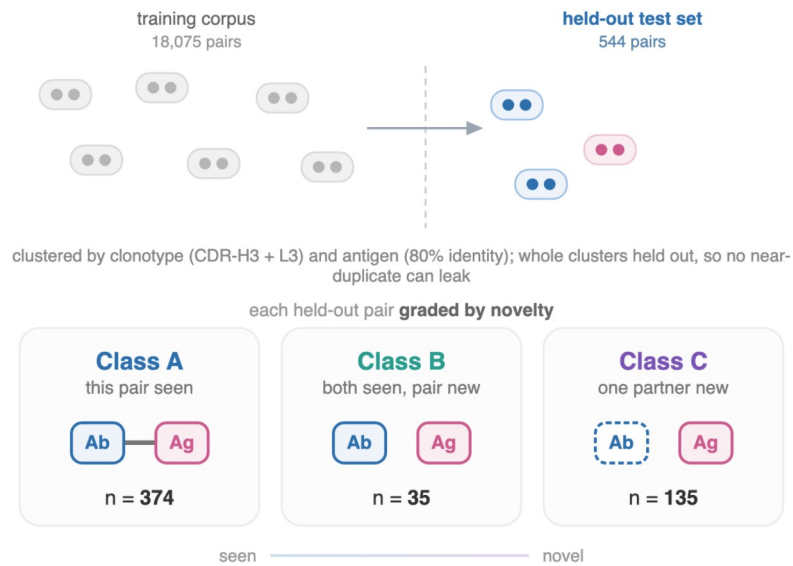
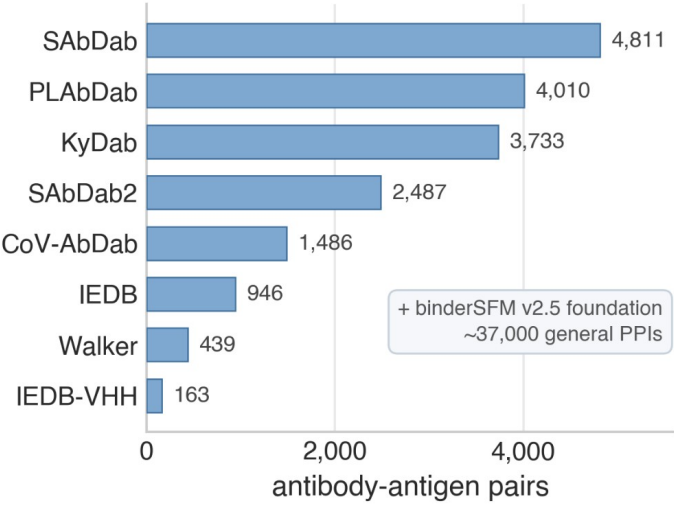
Contrastive learning for antibody-antigen sequence-to-specificity prediction

Hyunho Lee, Karla Castro, Sean Renwick, Lucas Stalder, Wiona Glänzer, Rachita Kumar, NingNing Chen, Andreas Scheck, Alex Yermanos, Derek Mason, Sai T. Reddy

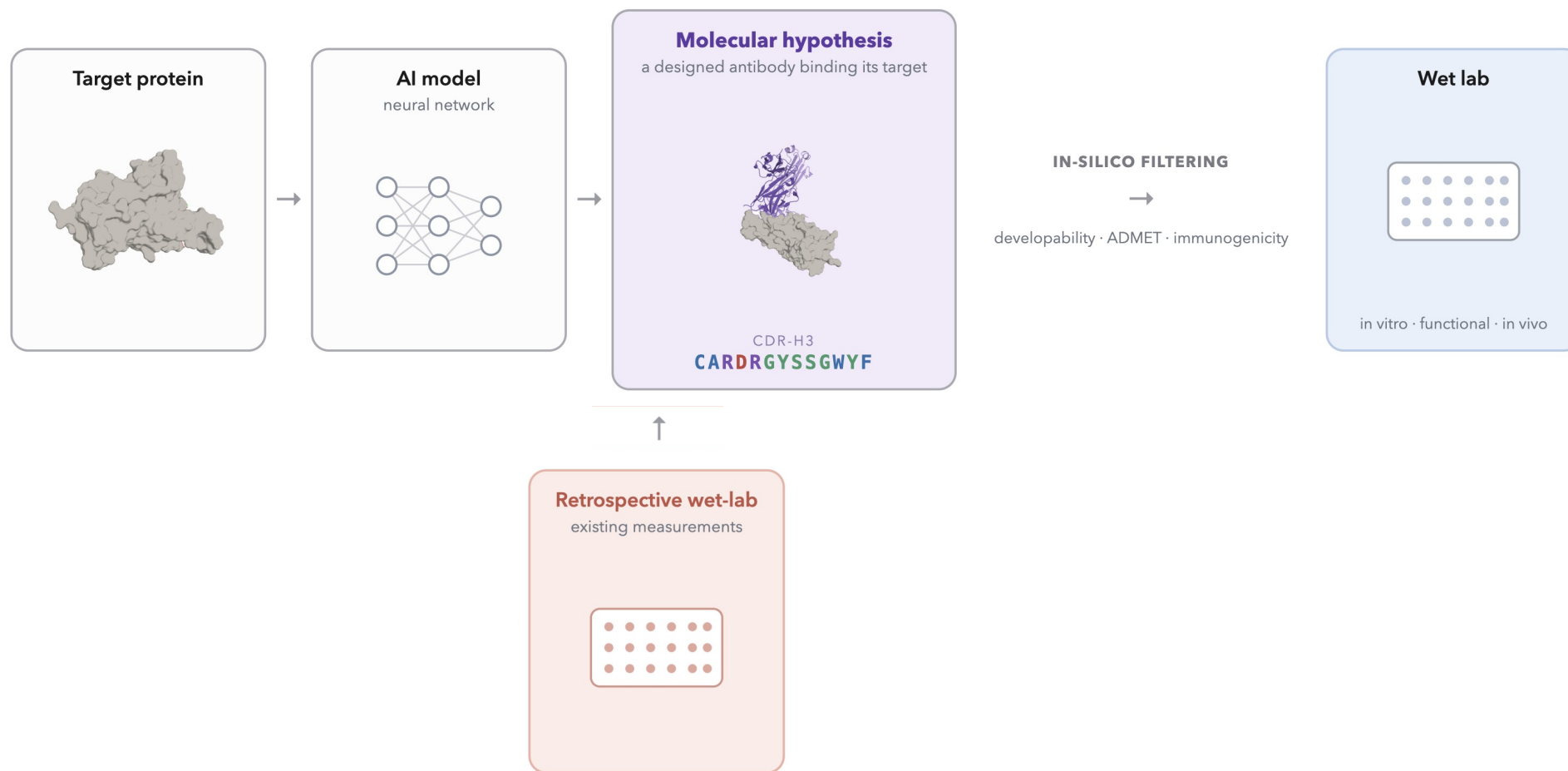
Antibody-Antigen Specificity Foundation Model (abSFM)



abSFM Performance Metrics



AI Drug Discovery is a Molecular Hypothesis



Retrospective Wet-lab Validation with abSFM



Diversity source

Natural – Shuffled VH/VL

- Diversity from healthy human donor
 - Up to 1e9 B cells per leukopak
 - Immune pre-selected
- All VH/VL frameworks combined
 - Codon-usage is human

Synthetic – CDR focused

- Diversity based on *In silico* design & DNA synthesis
- Up to 1e9 CDR3-only sequences
- Single VH/VL Framework
- Codon-usage can be adapted

Our Platforms

CHL display

- Full IgG display
- Mammalian PTM
- 8 hour doubling time
- Genomic libraries

Yeast display

- FAB display
- Yeast PTM
- 2 hour doubling time
- Genomic libraries or plasmid

334 BIIE-discovered mAbs across 5 antigens (RBD, RSV-F, CTLA-4, HA, RP009), each with a known cognate target

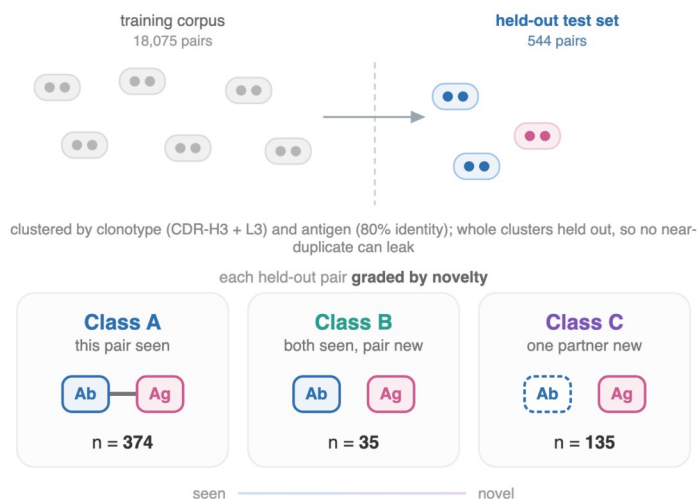
abSFM screen: scored every mAb against all ~5,700 training antigens – from sequence alone, no structure

Cognate-antigen z-score = how many SDs the true antigen scores above the whole ~5,700-antigen panel;

$z > 2$ = cognate in the top ~2% = "deorphanized"

All mAbs represent Class C

Leakage assessment of B1IE-mAbs with abSFM Train Data

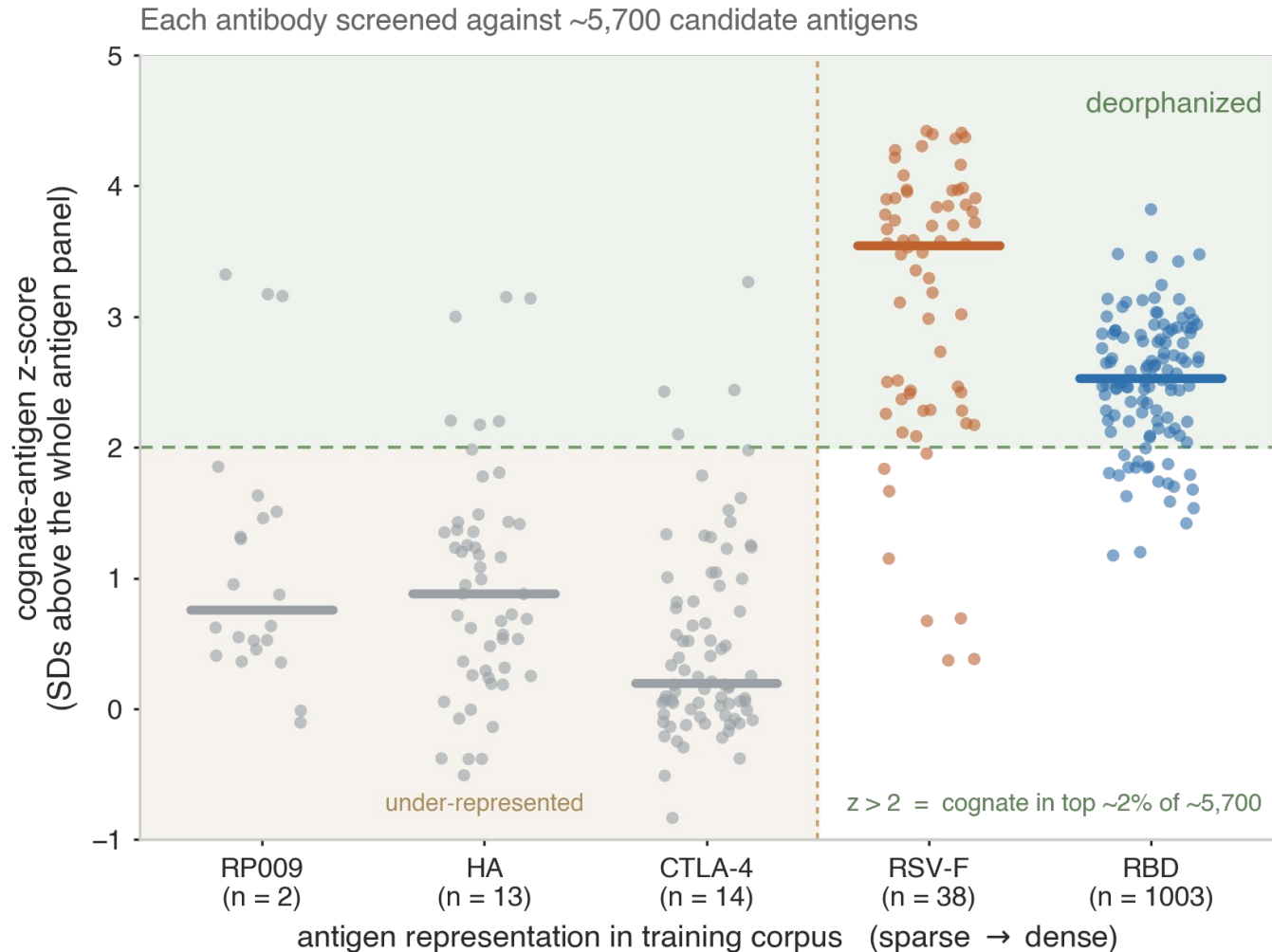


Sequence novelty of the repertoire-mining antibodies (per campaign) — all clones are Class C (antibody zero-shot); memorization ruled out

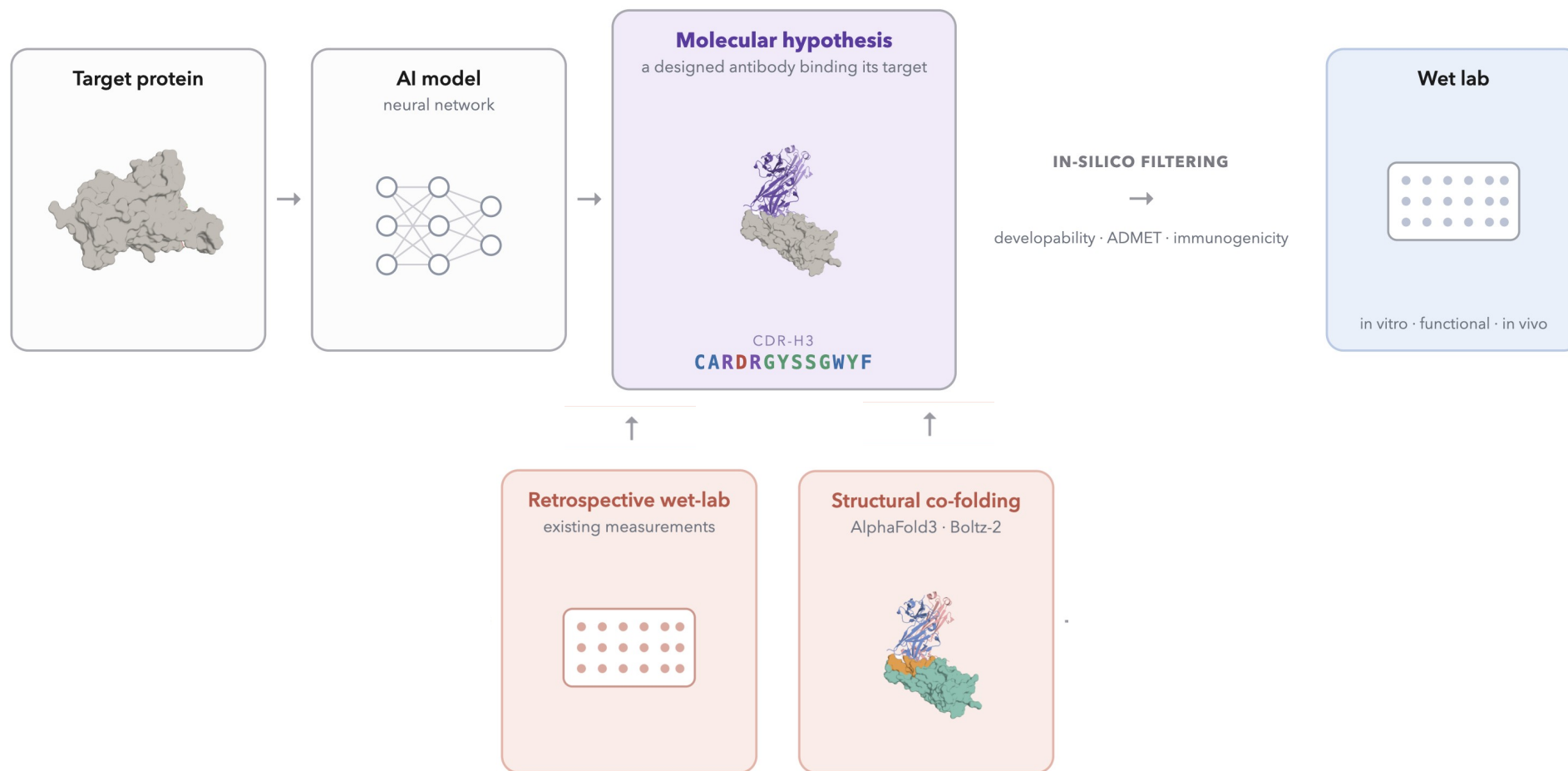
Campaign	Antigen	n antibodies	Class A	Class B	Class C	% Class C (zero-shot)	median nearest-train CDR-H3 identity
SC001	CoV2-RBD	53	0	0	53	100%	0.69
SC006	CoV2-RBD	60	0	0	60	100%	0.67
SC008	RSV-PreF	62	0	0	62	100%	0.64
SC002	hCTLA-4	34	0	0	34	100%	0.62
SC005	hCTLA-4	40	0	0	40	100%	0.53
SC009	Influenza-HA	51	0	0	51	100%	0.62
SC004	RP009	22	0	0	22	100%	0.65
Total	5 antigens	322	0	0	322	100%	0.64

Class C = antibody clonotype (paired CDR-H3 + CDR-L3, $\geq 80\%$ identity each) novel to the full training set; A = exact pair in training, B = clonotype seen with another antigen (Table S2). Nearest-train CDR-H3 identity is H3 only; no antibody reached $\geq 80\%$ on both CDR-H3 and CDR-L3, so none is Class A/B. 12 of 334 antibodies excluded (anchor-based CDR extraction failed).

abSFM Specificity Predictions Correlate with Train Data

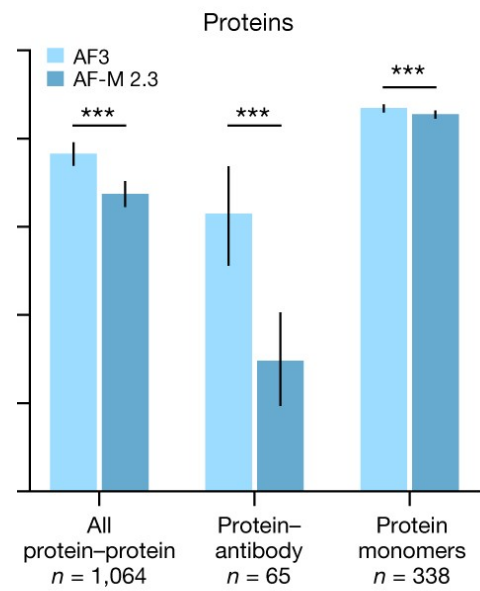
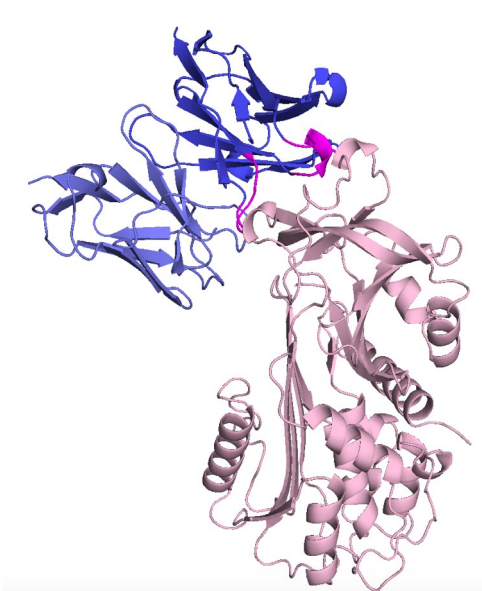


AI Drug Discovery is a Molecular Hypothesis

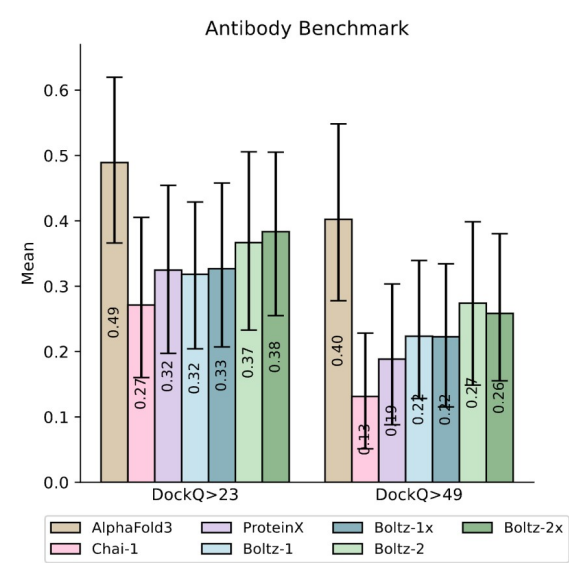


Structural Co-Folding Diffusion Models as Orthogonal Verifiers

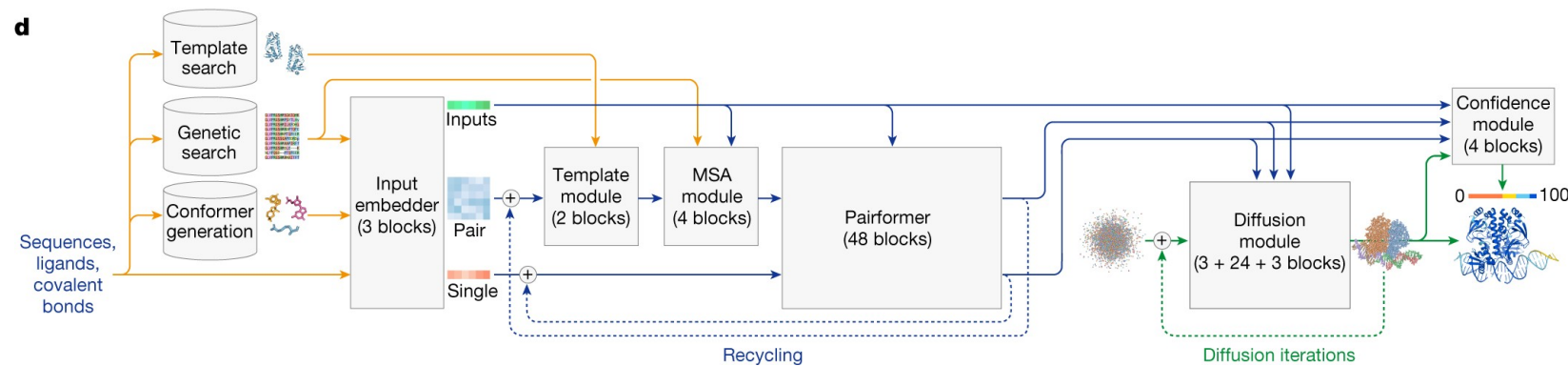
AlphaFold 3 and Boltz-2 offer valuable orthogonal verification



Abramson, Nature 2024.



Passaro, bioRxiv, 2025.



AF3 and Boltz-2 used for “FACS”

Co-folding Ab-Ag Pairs

AF3 (5 seeds)

Bz-2 (5 diff samples)

median per pair

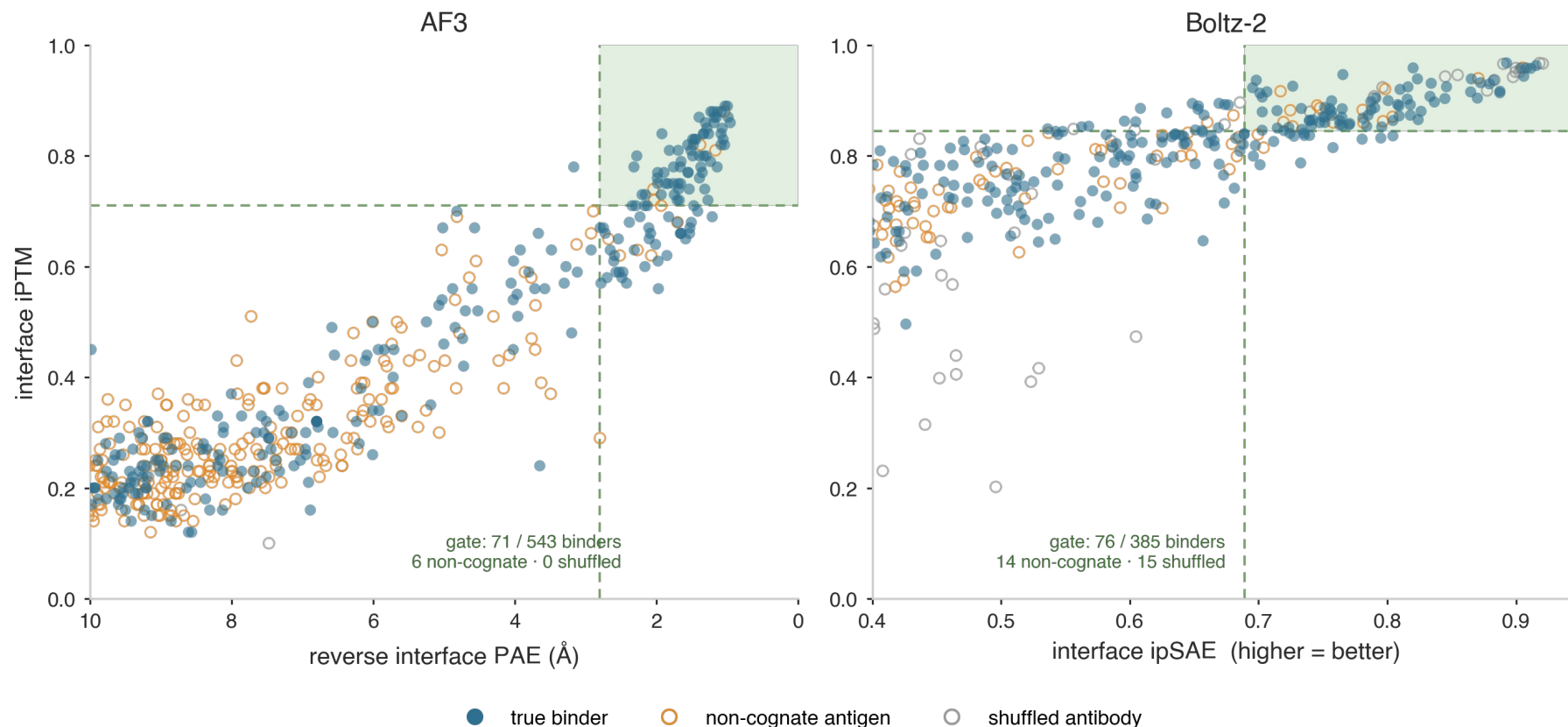
Cohort:

599 Ab-Ag (cognate)

597 (real Ab x wrong Ag)

547 shuffled Ab

(residue-shuff x cognate Ag)



iPTM (interface predicted TM-score): scores how well two docked molecules fit together (0 to 1, higher = better)

interface PAE (predicted-align error): expected position error in Ang, across interface (lower = better)

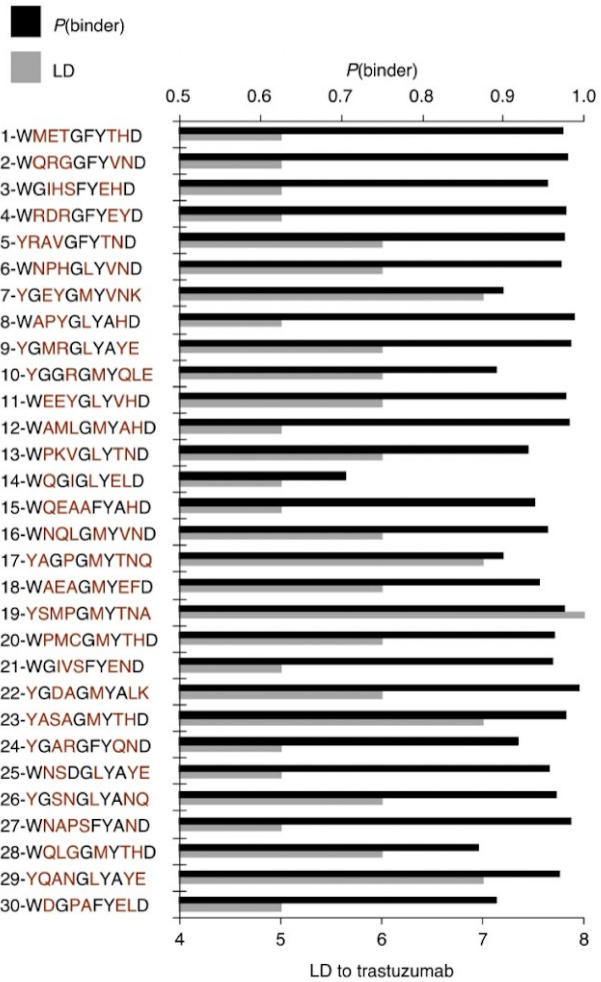
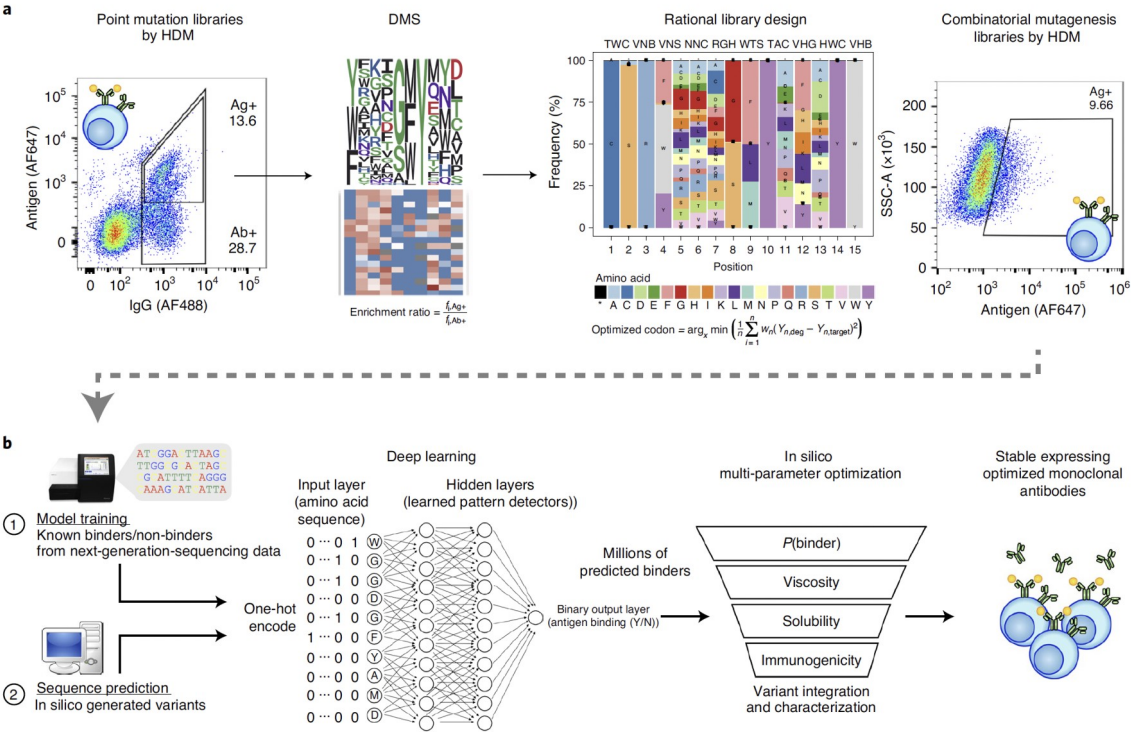
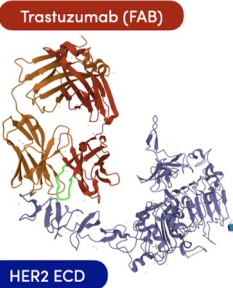
ipSAE (interface score from aligned error): interface PAE normalized (0 to 1, higher is better)

Epitope-specific Antibody Design by Combinatorial Mutagenesis

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Optimization of therapeutic antibodies by predicting antigen specificity from antibody sequence via deep learning

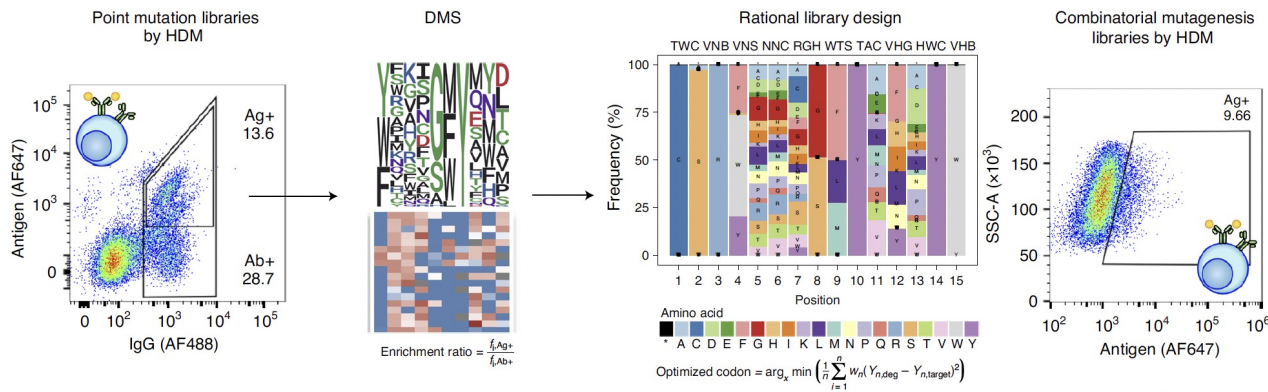
Derek M. Mason^{1,3}, Simon Friedensohn^{1,3}, Cédric R. Weber^{1,3}, Christian Jordi¹, Bastian Wagner¹, Simon M. Meng¹, Roy A. Ehling¹, Lucia Bonati¹, Jan Dahinden¹, Pablo Gainza², Bruno E. Correia² and Sai T. Reddy^{1,3}



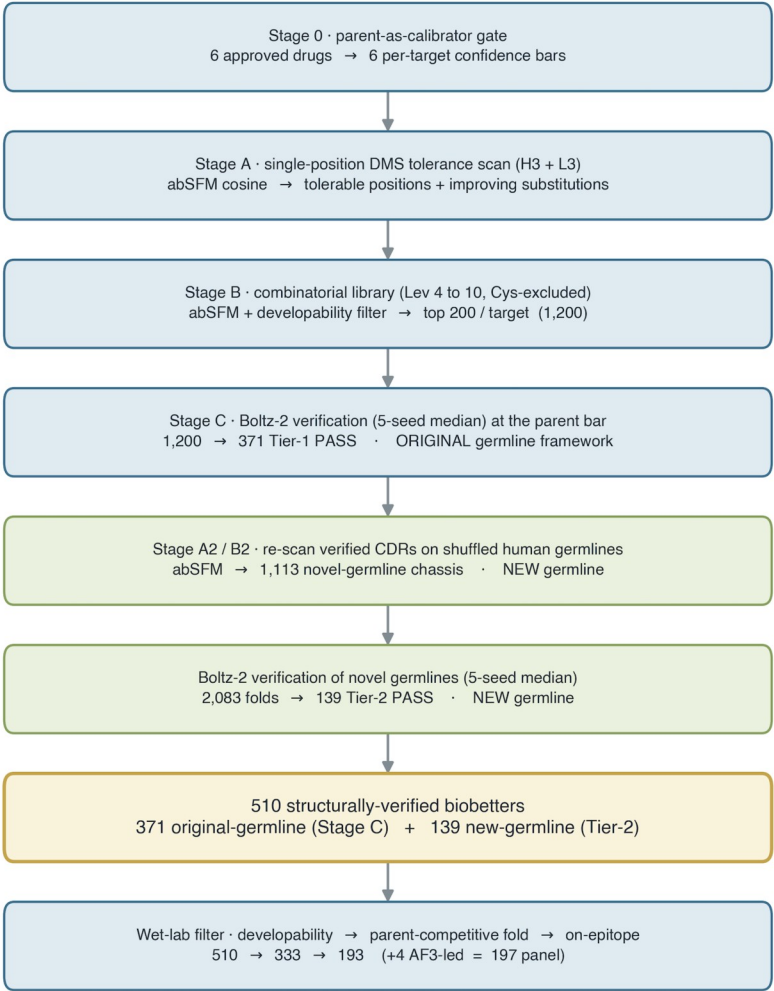
Generative Antibody Design in a Loop with abSFM (abGenLoop)

Clinical parent antibodies — CDR-H3 / CDR-L3 reference (each design is benchmarked against its own parent)

Target	Parent (drug)	CDR-H3	CDR-L3
PD-L1	atezolizumab	RHWPGGFDY	QQYLYHPAT
IgE	omalizumab	GSHYFGHWHFAV	QQSHEDPYT
TNF α	adalimumab	VSYLSTASSLDY	QRYNRAPYT
RSV-F IV	clesrovimab	GAPYGGNSDYGYGLDV	QQFLDFPFT
RSV-F $\emptyset$	nirsevimab	ETALVVSETYLPHYFDN	QQYDNLPLT
PCSK9	alirocumab	DSNWGNFDL	QQYYTTPYT



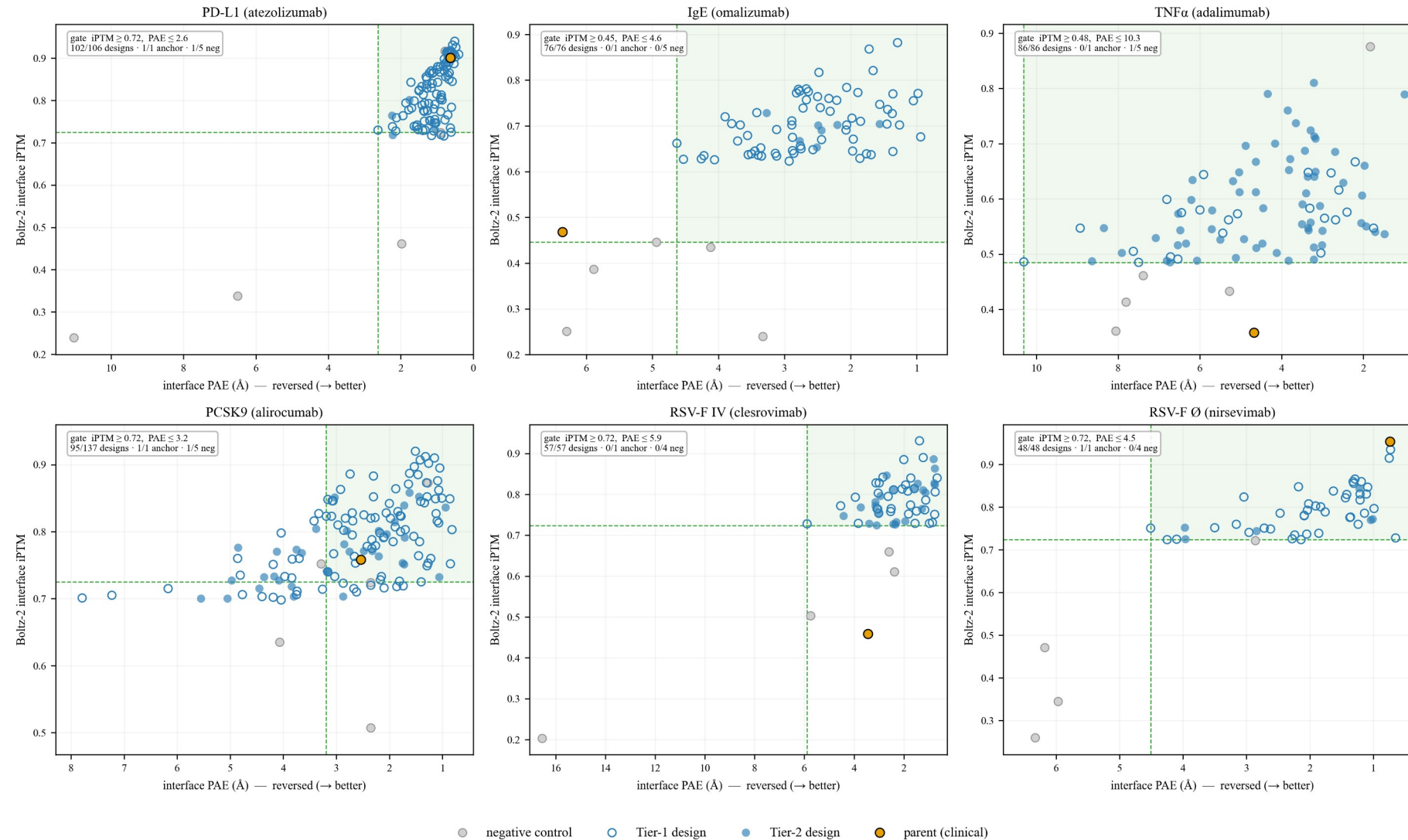
abGenLoop — campaign funnel (6 clinical parent antibodies)



Linear pipeline: new germlines are grafted only after the Stage C verify; abSFM proposes, Boltz-2 / AF3 dispose.

Structural Verification of abGenLoop variants

Per-target binding verification — Boltz-2 (5-seed median): designs, parent anchor, negative controls, FACS-style binder gate

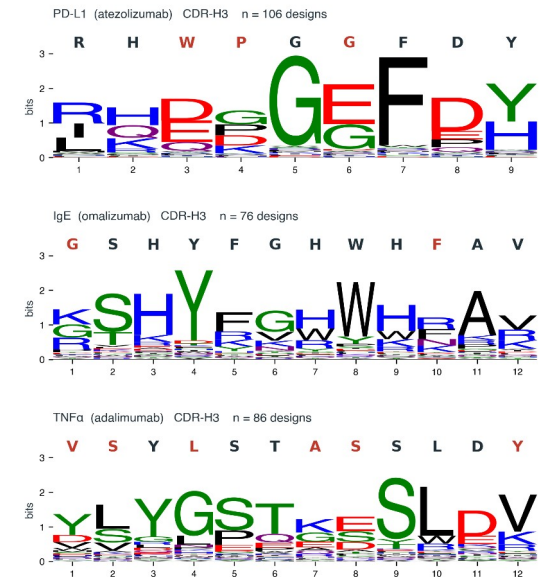


abGenLoop Final Candidates for Wet-lab testing

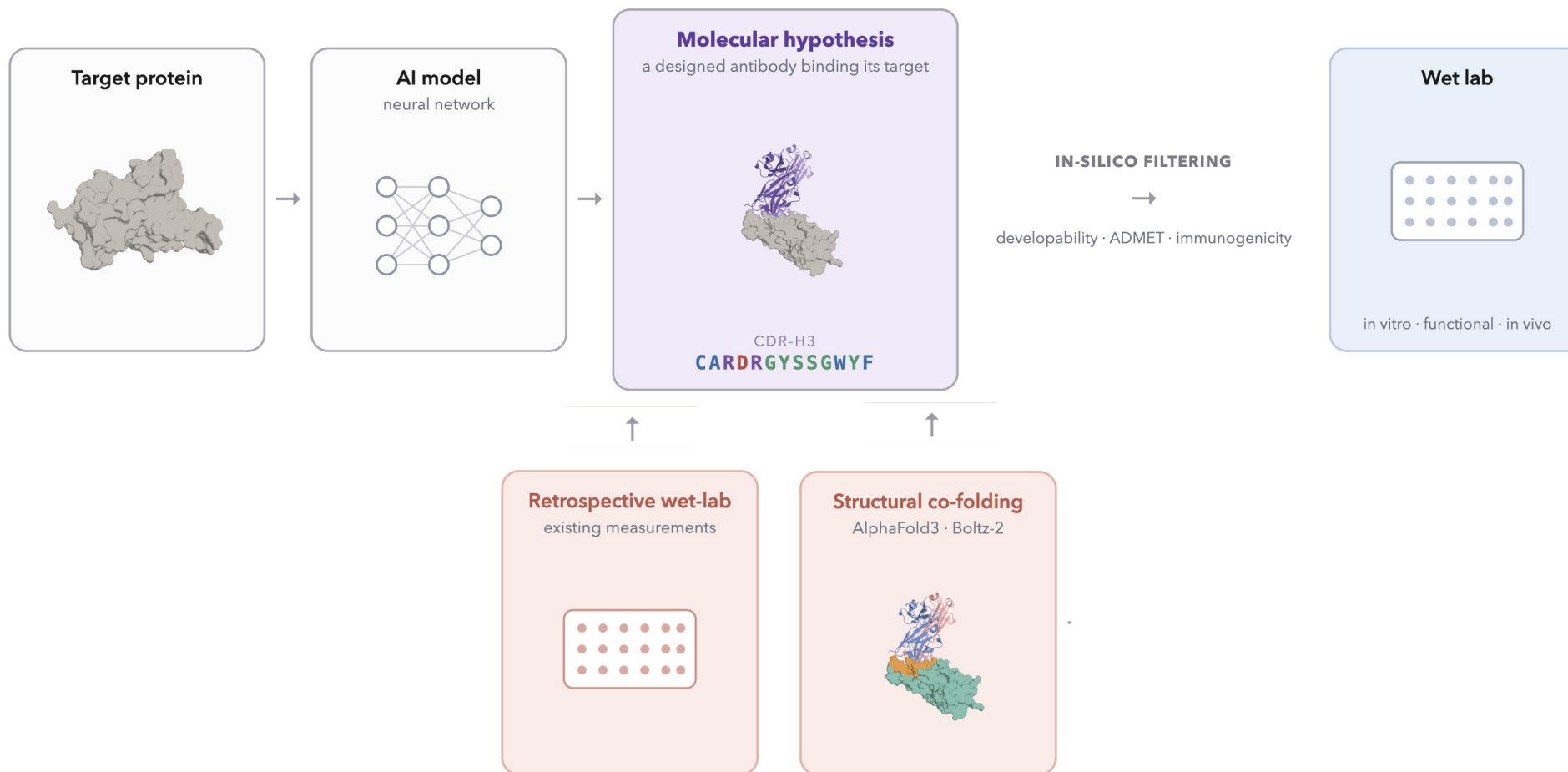
Per-target biobetters (Boltz-2, normalized to each drug) · $\geq 1.0\times$ = folds/interfaces at least as well as the approved antibody

Target (clinical parent)	Original-GL biobetters	New-GL biobetters	iPTM ÷ parent (median)	PAE ÷ parent (median)	epitope-confirmed	median H3 edits
PD-L1 (atezolizumab)	101	5	0.92x	0.64x	106 / 106	4
IgE (omalizumab)	69	7	1.50x	2.40x	74 / 76	5
TNF α (adalimumab)	23	63	1.60x	1.18x	64 / 86	7
PCSK9 (alirocumab)	102	35	1.04x	1.09x	9 / 137*	4
RSV-F IV (clesrovimab)	34	23	1.71x	1.67x	26 / 57	7
RSV-F Ø (nirsevimab)	42	6	0.82x	0.40x	19 / 48	7
All 6 targets	371	139	510 verified		298 / 510	

* PCSK9 epitope readout unreliable (Boltz-2 fold-quality, not real epitope drift); read its iPTM relative to parent.



AI Drug Discovery is a Molecular Hypothesis

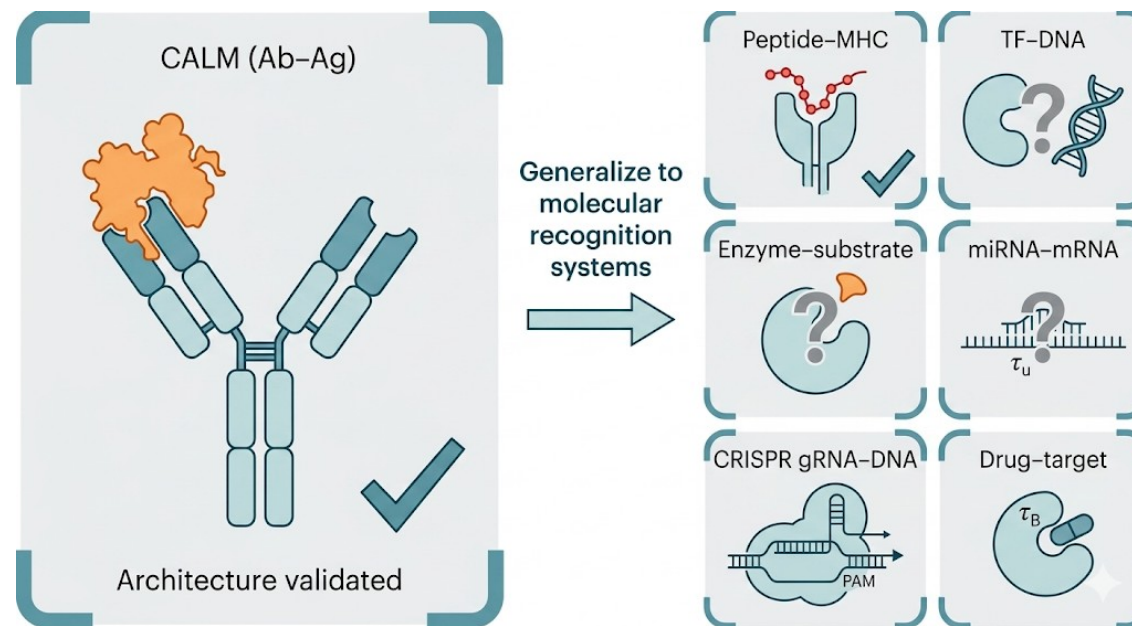
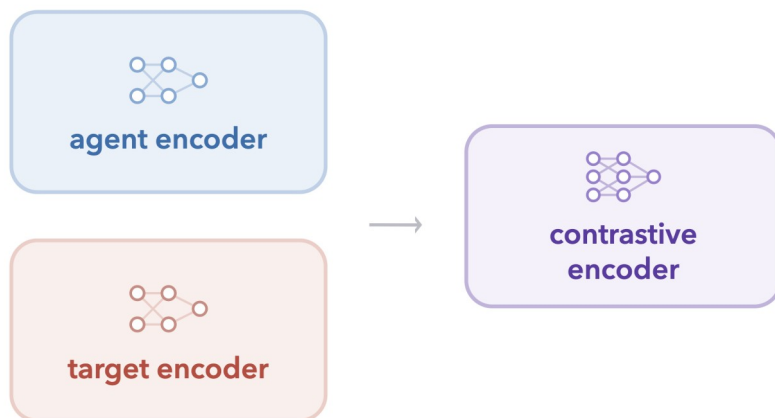


Specificity Foundation Models (SFM)

A new class of Sequence-to-Specificity Foundation Models

SFM Architecture

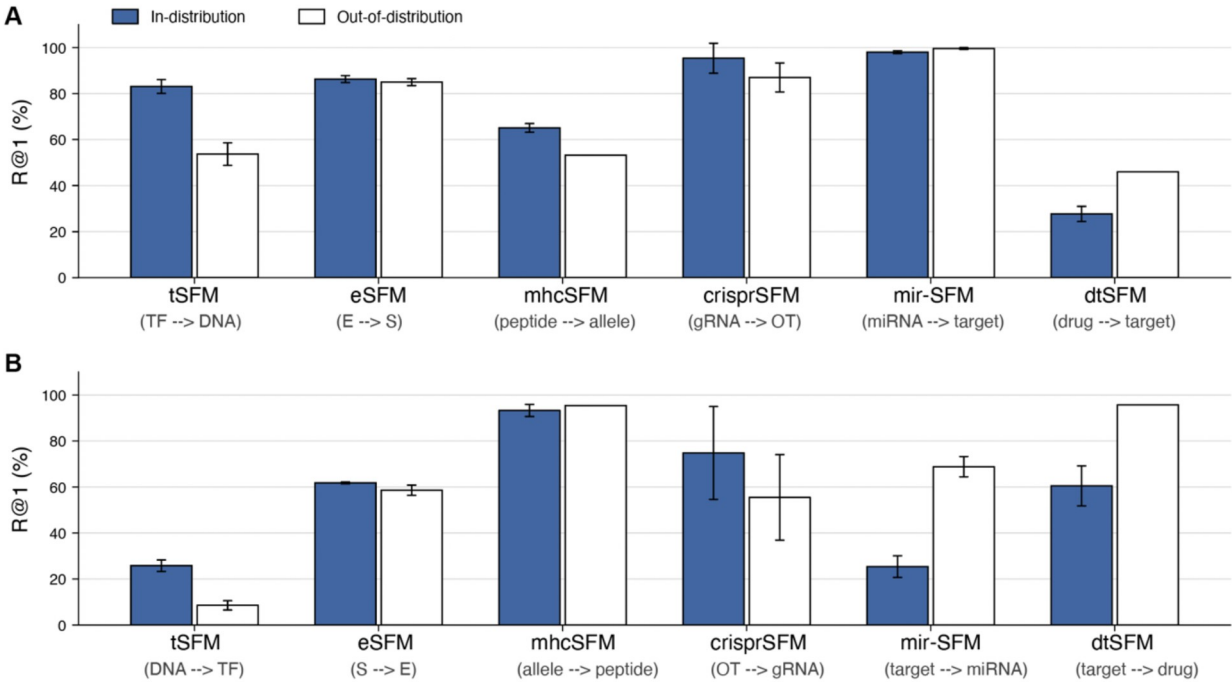
If architecture is correct, can transfer across other molecular recognition systems



Cross-domain Performance of SFMs

Prototypes with training data, R@1 in pool 512 (random 0.2%)

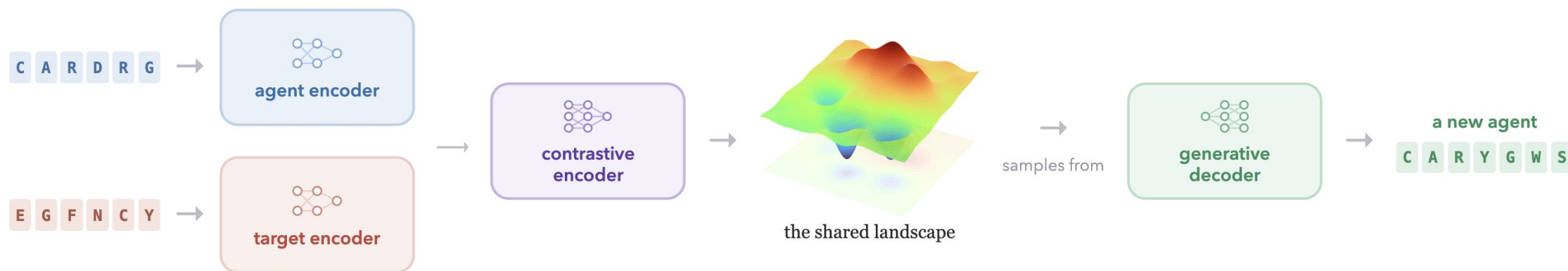
SFM	Domain	Data source	Pairs	Unique agents / targets	Encoders (agent / target)	Bilinearity
tSFM	TF–DNA	JASPAR 2022	20,540	1,060 TFs / 15,425 DNA motifs	ESM-2 / DNABERT-2	Given
eSFM	Enzyme–substrate	ReactZyme	177,442	177,389 enzymes / 2,646 substrates	ESM-2 / MoLFormer-XL	Emergent
mhcSFM	Peptide–MHC I	NetMHCpan 4.1 BA	168,710	38,837 peptides / 105 alleles	ESM-2 / ESM-2	Given
crisprSFM	gRNA–genomic OT	CCLMoff	30,342	1,668 guides / 30,308 OT sites	DNABERT-2 / DNABERT-2	Given
mir-SFM	miRNA–mRNA	ENCORI CLIP-seq	206,692	135 miRNAs / 110,221 targets	DNABERT-2 / DNABERT-2	Given
dtSFM	Drug–target	DAVIS / KIBA / BindingDB	121,548	10,338 drugs / 1,358 targets	MoLFormer-XL / ESM-2	Emergent



SFMs are also Generative Molecular Design Engines

Specificity Foundation Models

Recognition and design from a single shared landscape



Scaling up the small molecule drug-target protein SFM (dtSFM)

Largest public available training dataset

dtSFM v3

715,00 drug-target pairs

522,775 drugs

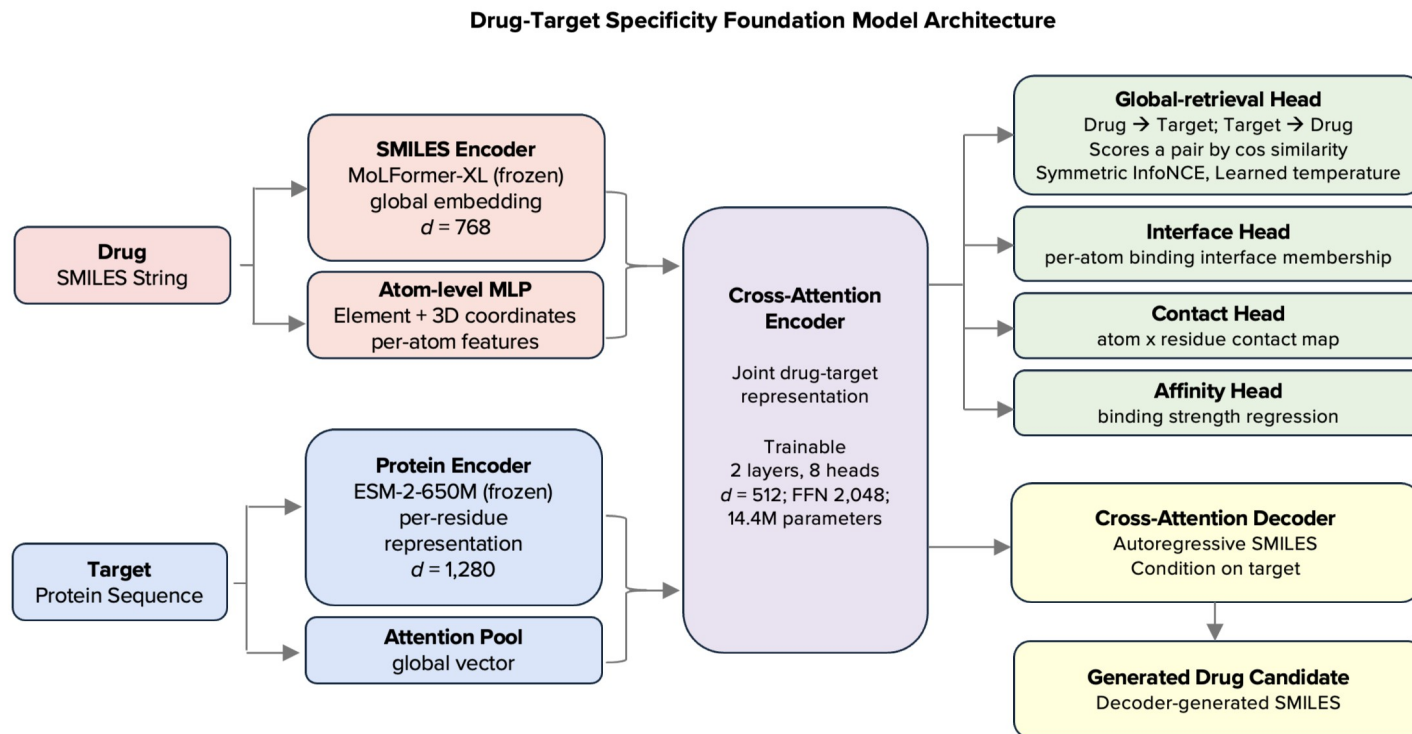
22,964 proteins (proteome-scale)

Generative design

16 targets immune/oncology

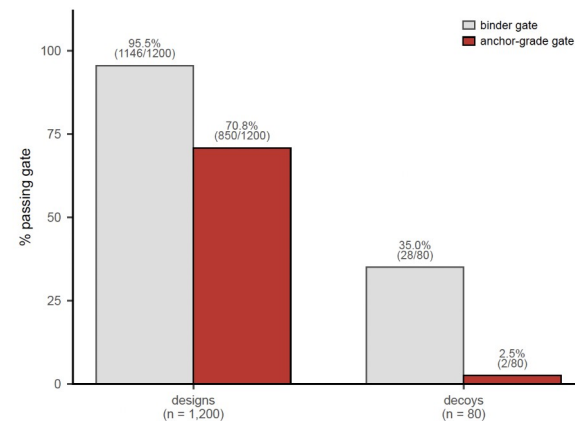
1,200 total designs

AF3 Structural Co-folding for verification



dtSFM Generated Designs are Structurally Verified by AF3

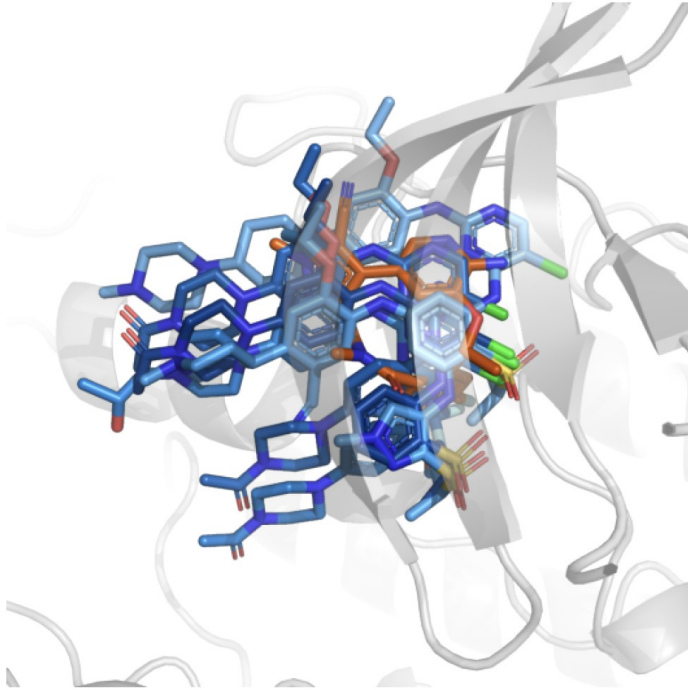
Show high quality molecular properties of drug development



Orthogonal Verification with Structural Co-Folding

AlphaFold 3 does not have correlated scores with dtSFM

dtSFM Generated Variants Target Same Sites of Clinical Anchors



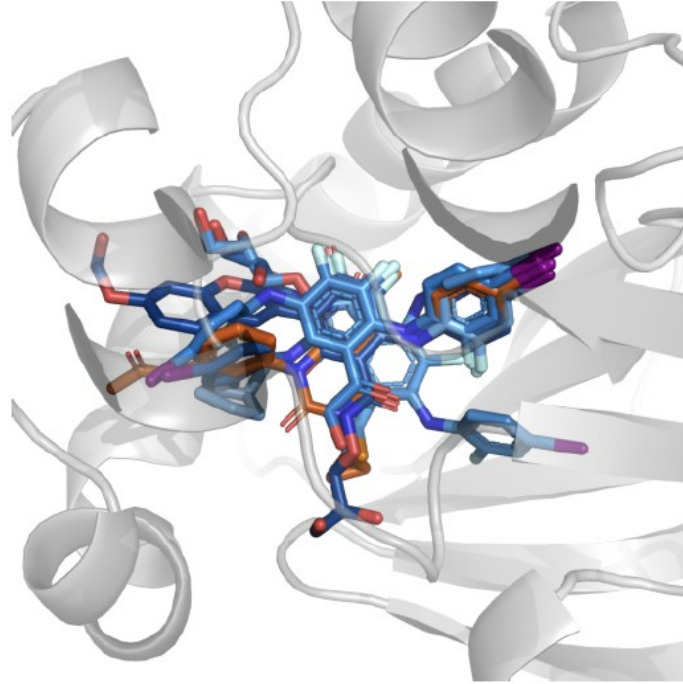
ALK

+ lorlatinib

d0063, d0045, d0053, d0034, d0062

mean iPTM = 0.98 mean PAE = 0.9 Å

mean Tanimoto to lorlatinib = 0.12



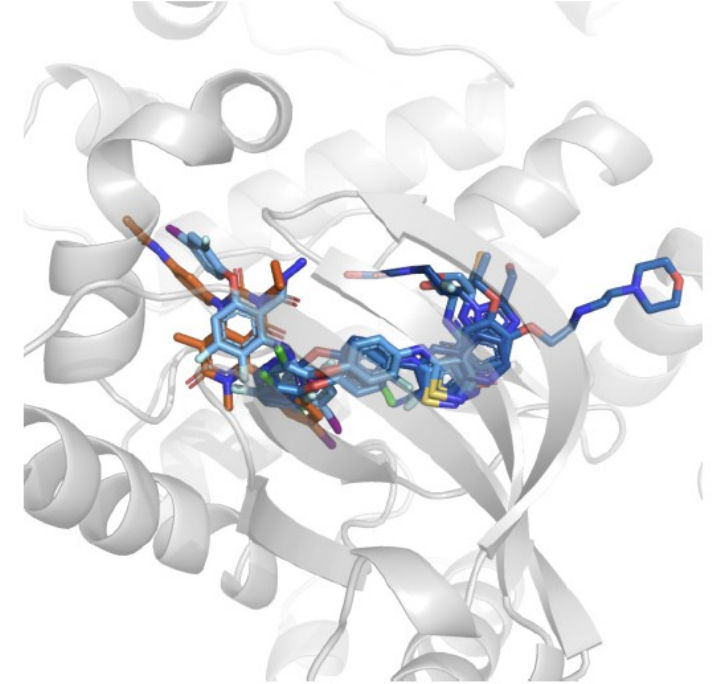
MAP2K1

+ trametinib

d0805, d0806, d0808, d0830, d0859

mean iPTM = 0.88 mean PAE = 1.8 Å

mean Tanimoto to trametinib = 0.27



MAP2K2

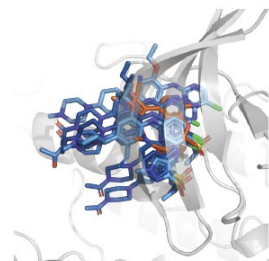
+ trametinib

d0938, d0895, d0913, d0933, d0878

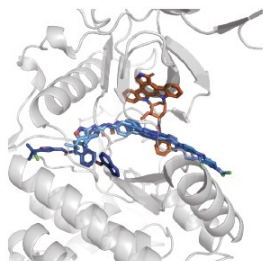
mean iPTM = 0.97 mean PAE = 0.8 Å

mean Tanimoto to trametinib = 0.16

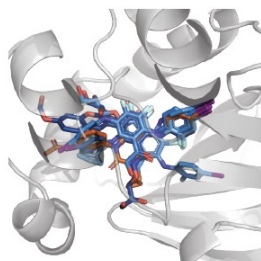
dtSFM Generated Variants Target Same Sites of Clinical Anchors



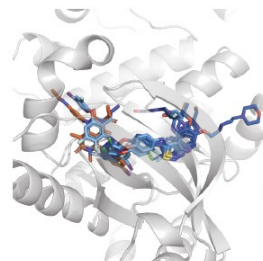
ALK + lorlatinib
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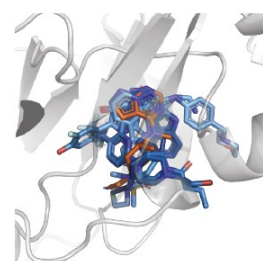
FLT3 + midostaurin
d0467, d0477, d0458, d0487, d0486
mean IPTM = 0.98 mean PAE = 0.9 Å
mean Tanimoto to midostaurin = 0.09



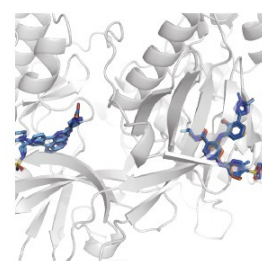
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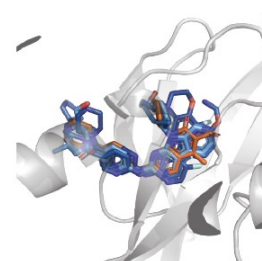
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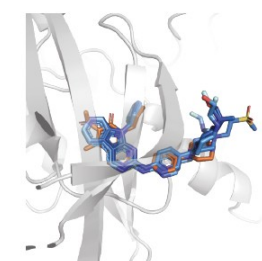
JAK3 + tofacitinib
d0761, d0722, d0776, d0735, d0767
mean IPTM = 0.98 mean PAE = 0.9 Å
mean Tanimoto to tofacitinib = 0.16



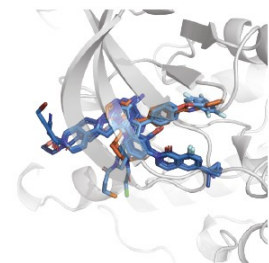
TYK2 + deucravacitinib
d1234, d1247, d1287, d1270, d1266
mean IPTM = 0.95 mean PAE = 1.6 Å
mean Tanimoto to deucravacitinib = 0.27



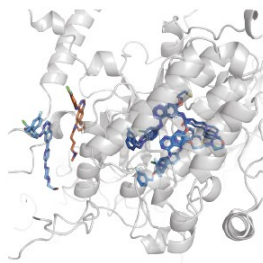
CDK4 + palbociclib
d0216, d0173, d0217, d0218, d0161
mean IPTM = 0.94 mean PAE = 0.9 Å
mean Tanimoto to palbociclib = 0.33



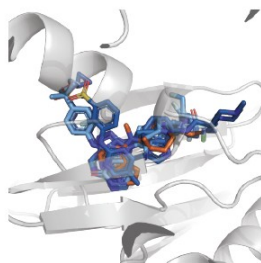
CDK6 + palbociclib
d0231, d0243, d0262, d0270, d0283
mean IPTM = 0.95 mean PAE = 0.8 Å
mean Tanimoto to palbociclib = 0.37



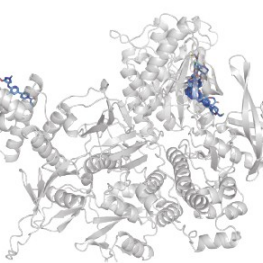
BTK + acalabrutinib
d0116, d0151, d0145, d0073, d0076
mean IPTM = 0.97 mean PAE = 0.8 Å
mean Tanimoto to acalabrutinib = 0.18



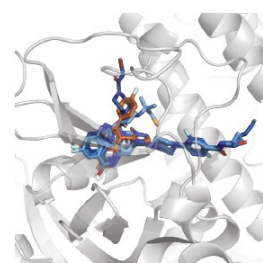
EGFR + afatinib
d0335, d0339, d0355, d0309, d0296
mean IPTM = 0.98 mean PAE = 1.0 Å
mean Tanimoto to afatinib = 0.05



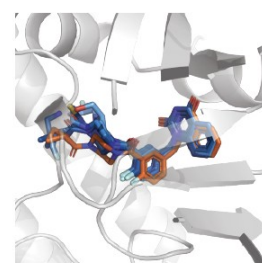
JAK1 + tofacitinib
d0572, d0596, d0612, d0586, d0590
mean IPTM = 0.96 mean PAE = 1.2 Å
mean Tanimoto to tofacitinib = 0.21



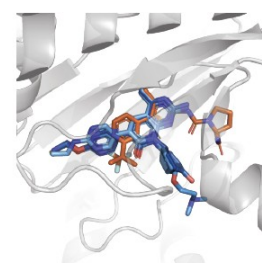
JAK2 + tofacitinib
d0652, d0639, d0633, d0634, d0635
mean IPTM = 0.90 mean PAE = 2.8 Å
mean Tanimoto to tofacitinib = 0.15



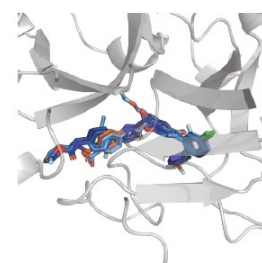
PARP1 + talazoparib
d1029, d1031, d1001, d1002, d1003
mean IPTM = 0.96 mean PAE = 1.0 Å
mean Tanimoto to talazoparib = 0.15



PARP2 + olaparib
d1049, d1062, d1064, d1066, d1069
mean IPTM = 0.98 mean PAE = 0.8 Å
mean Tanimoto to olaparib = 0.43

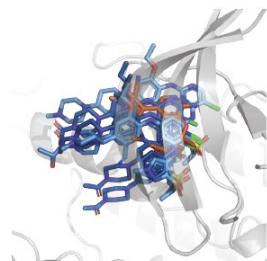


PIK3CA + alpelisib
d1168, d1201, d1204, d1141, d1150
mean IPTM = 0.98 mean PAE = 0.8 Å
mean Tanimoto to alpelisib = 0.18

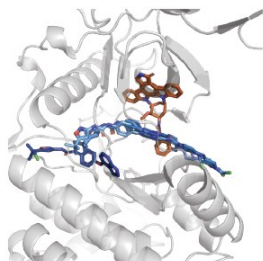


F11 + asundexian
d0417, d0433, d0390, d0393, d0408
mean IPTM = 0.98 mean PAE = 0.8 Å
mean Tanimoto to asundexian = 0.23

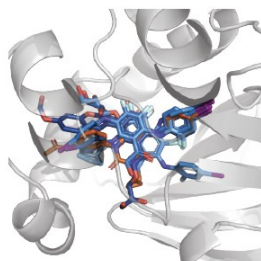
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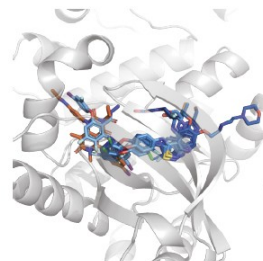
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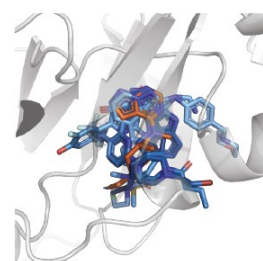
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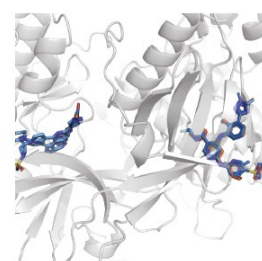
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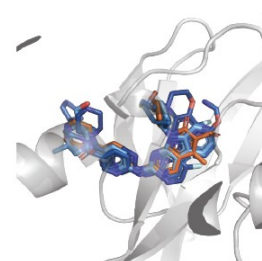
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mean IPTM = 0.97 mean PAE = 0.8 Å
mean Tanimoto to trametinib = 0.16



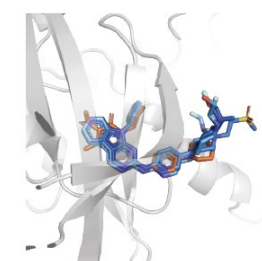
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d0761, d0722, d0776, d0735, d0767
mean IPTM = 0.98 mean PAE = 0.9 Å
mean Tanimoto to tofacitinib = 0.16



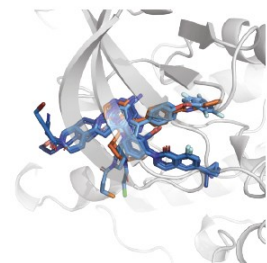
TYK2 + deucravacitinib
d1234, d1247, d1287, d1270, d1266
mean IPTM = 0.95 mean PAE = 1.6 Å
mean Tanimoto to deucravacitinib = 0.27



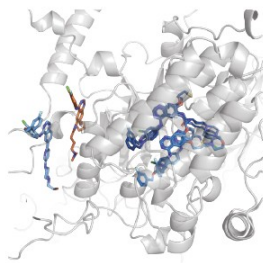
CDK4 + palbociclib
d0216, d0173, d0217, d0218, d0161
mean IPTM = 0.94 mean PAE = 0.9 Å
mean Tanimoto to palbociclib = 0.33



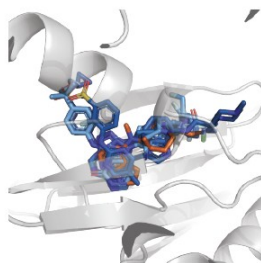
CDK6 + palbociclib
d0231, d0243, d0262, d0270, d0283
mean IPTM = 0.95 mean PAE = 0.8 Å
mean Tanimoto to palbociclib = 0.37



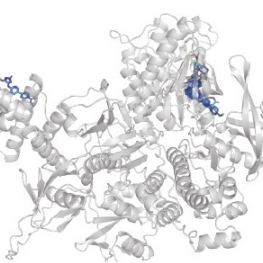
BTK + acalabrutinib
d0116, d0151, d0145, d0073, d0076
mean IPTM = 0.97 mean PAE = 0.8 Å
mean Tanimoto to acalabrutinib = 0.18



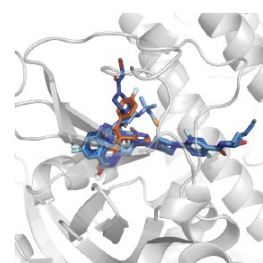
EGFR + afatinib
d0335, d0339, d0355, d0309, d0296
mean IPTM = 0.98 mean PAE = 1.0 Å
mean Tanimoto to afatinib = 0.05



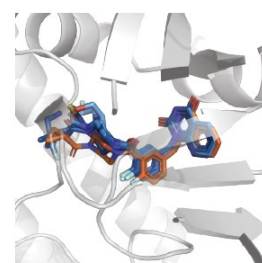
JAK1 + tofacitinib
d0572, d0596, d0612, d0586, d0590
mean IPTM = 0.96 mean PAE = 1.2 Å
mean Tanimoto to tofacitinib = 0.21



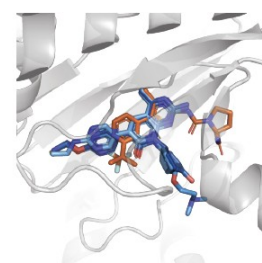
JAK2 + tofacitinib
d0652, d0639, d0633, d0634, d0635
mean IPTM = 0.90 mean PAE = 2.8 Å
mean Tanimoto to tofacitinib = 0.15



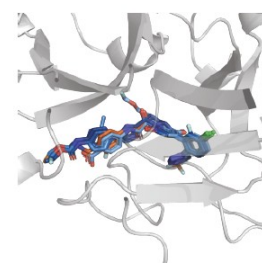
PARP1 + talazoparib
d1029, d1031, d1001, d1002, d1003
mean IPTM = 0.96 mean PAE = 1.0 Å
mean Tanimoto to talazoparib = 0.15



PARP2 + olaparib
d1049, d1062, d1064, d1066, d1069
mean IPTM = 0.98 mean PAE = 0.8 Å
mean Tanimoto to olaparib = 0.43



PIK3CA + alpelisib
d1168, d1201, d1204, d1141, d1150
mean IPTM = 0.98 mean PAE = 0.8 Å
mean Tanimoto to alpelisib = 0.18



F11 + asundexian
d0417, d0433, d0390, d0393, d0408
mean IPTM = 0.98 mean PAE = 0.8 Å
mean Tanimoto to asundexian = 0.23

dtSFM Generated Variants for GLP1 receptor family

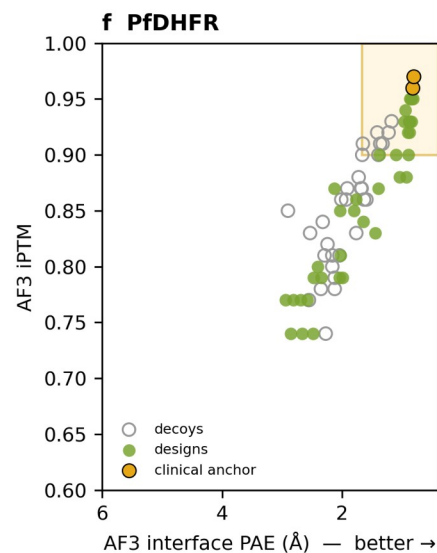
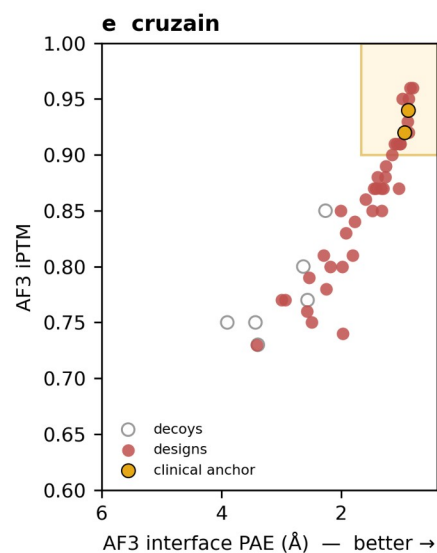
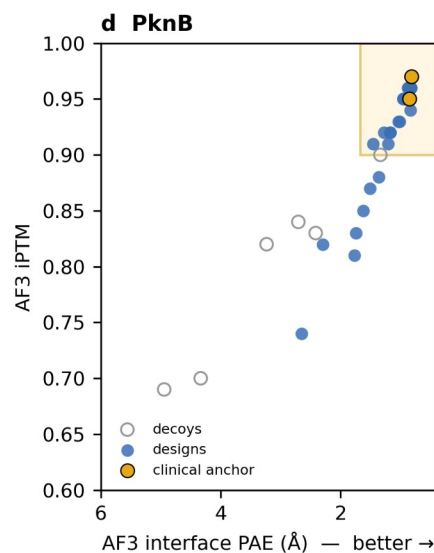
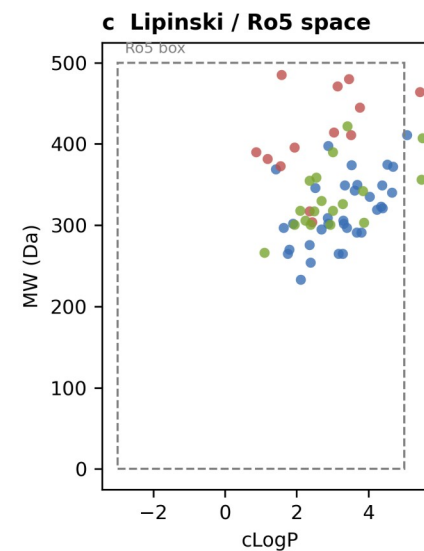
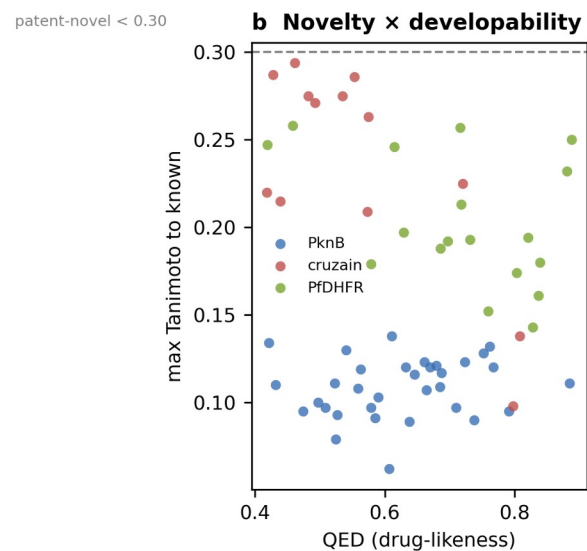
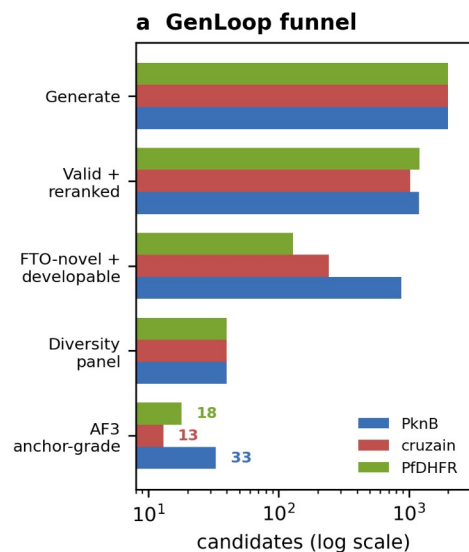
GenLoop pipeline on Global Health Targets

Target	Disease (pathogen)	Burden (WHO)	Current drugs	Key limitation / gap	Chemical / patent white-space
PknB	Tuberculosis (<i>M. tuberculosis</i>)	~1.25 M deaths/yr	first-line regimen; no PknB-directed drug	essential kinase; no clinical inhibitor; human-kinome selectivity unsolved	open; sparse chemistry (~120 ChEMBL)
Cruzain	Chagas disease (<i>T. cruzi</i>)	6–7 M infected; ~12,000 deaths/yr	benznidazole, nifurtimox	toxic; poor chronic cure; K777 discontinued (~2013)	validated target, innovation gap
PfDHFR	Malaria (<i>P. falciparum</i>)	~600,000 deaths/yr	pyrimethamine + sulfadoxine	quadruple-mutant resistance; P218 sets the bar	2,4-diaminopyrimidine patent expired Feb 2024

Targets: PknB (tuberculosis) · cruzain (Chagas) · PfDHFR (malaria)

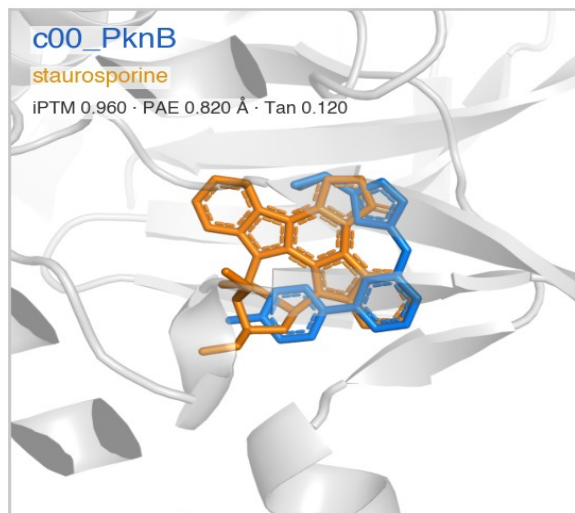


GenLoop funnel and per-target binding verification

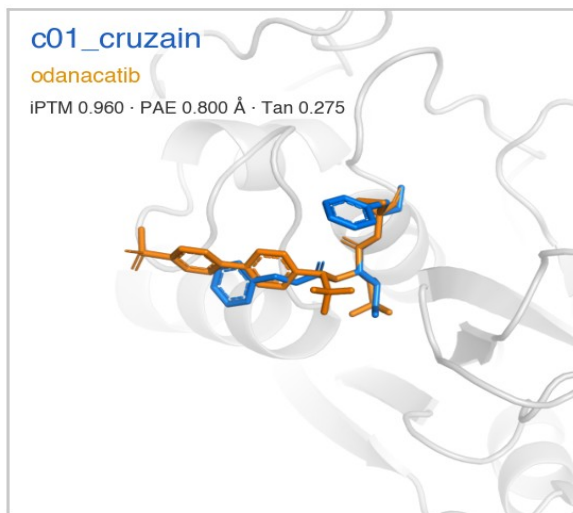


Generated Variants show High AF3 Confidence

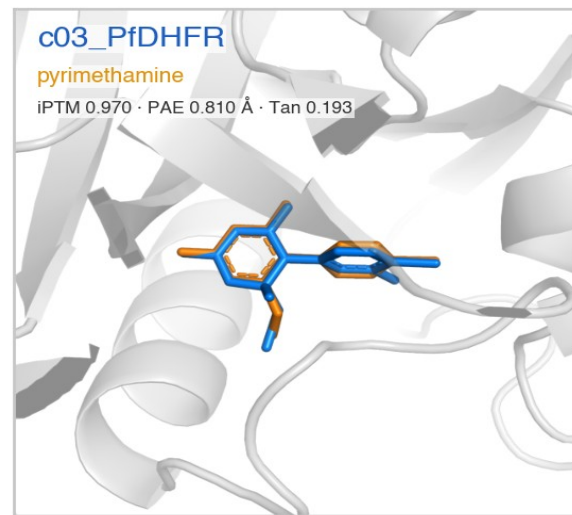
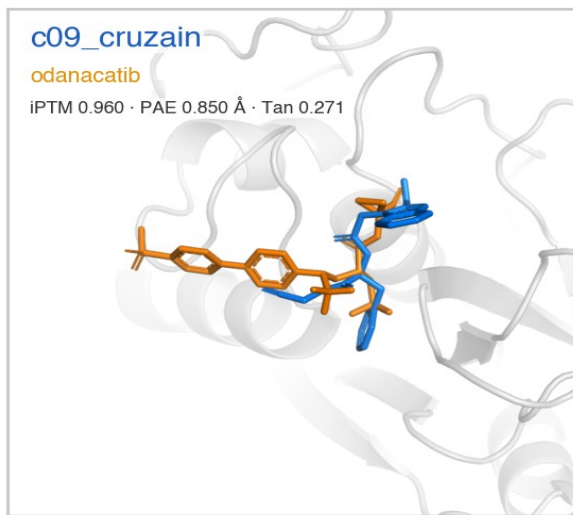
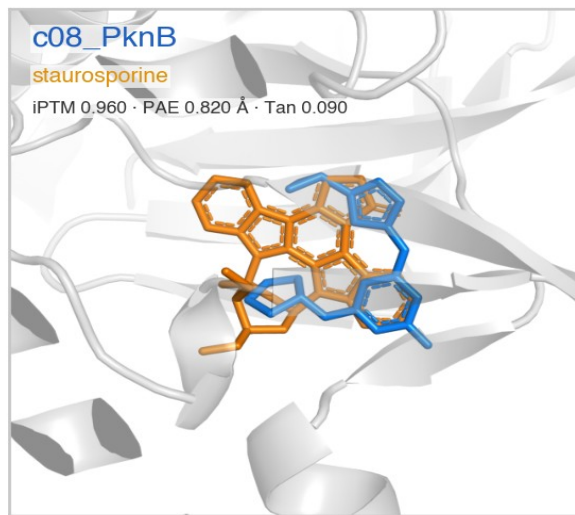
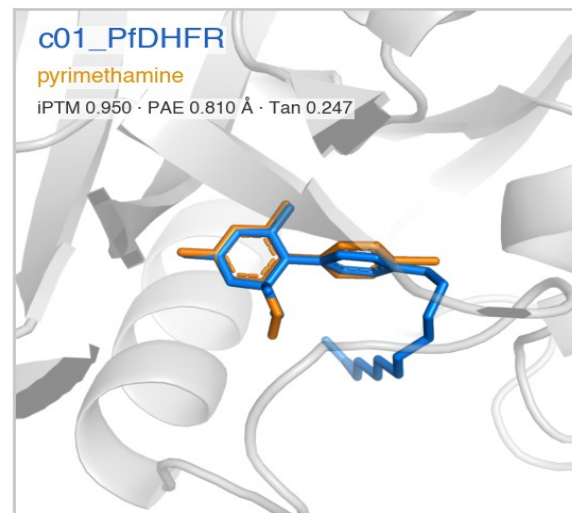
PknB



cruzain

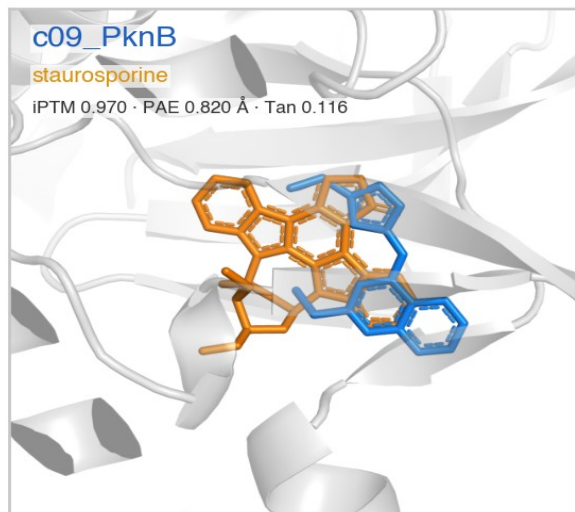


PfDHFR

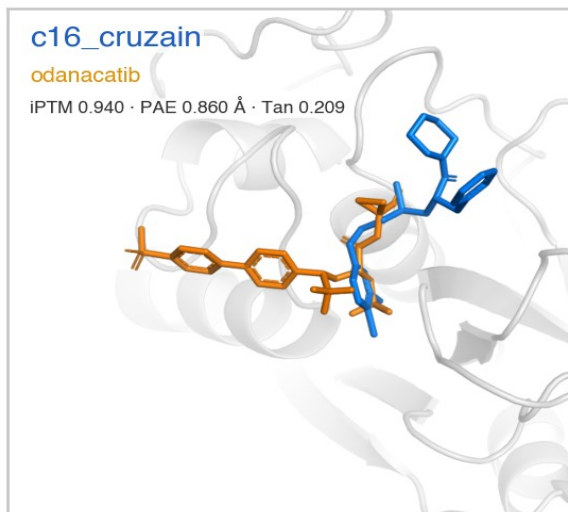


Generated Variants show High AF3 Confidence

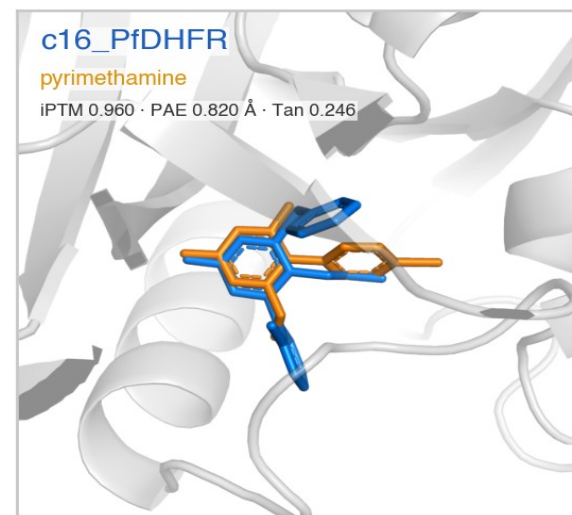
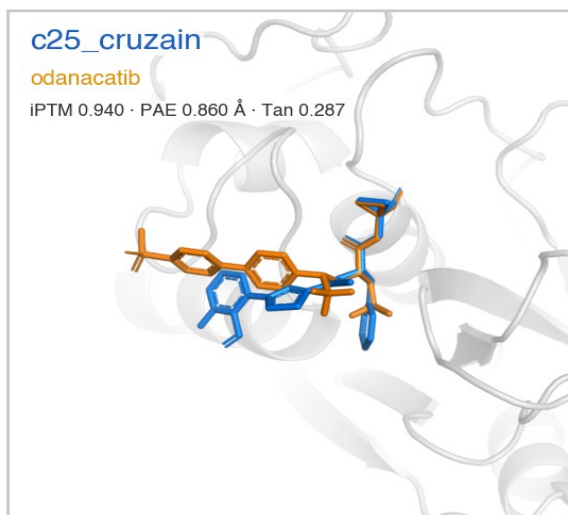
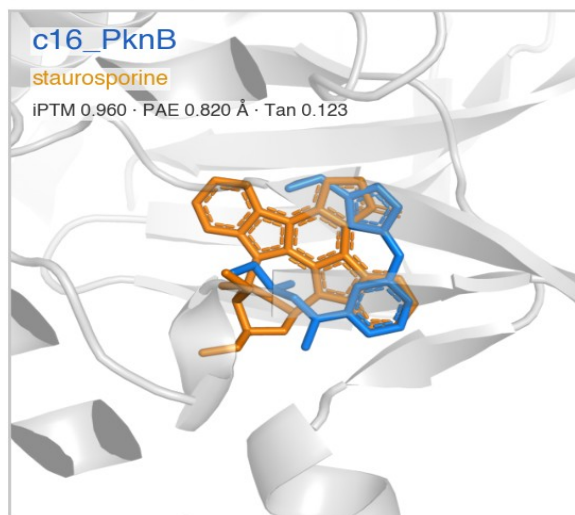
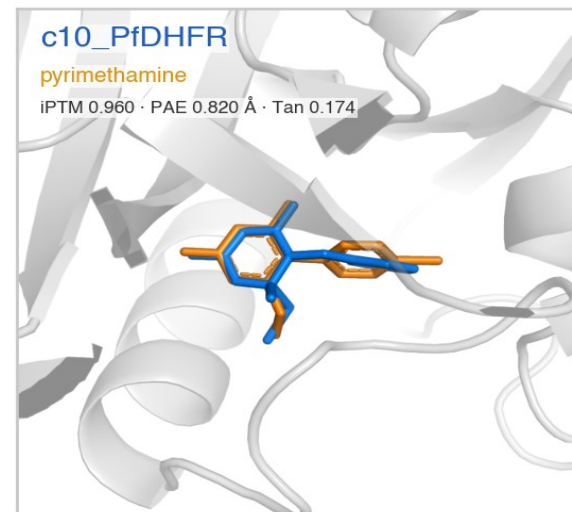
PknB



cruzain



PfDHFR



dtSFM v3

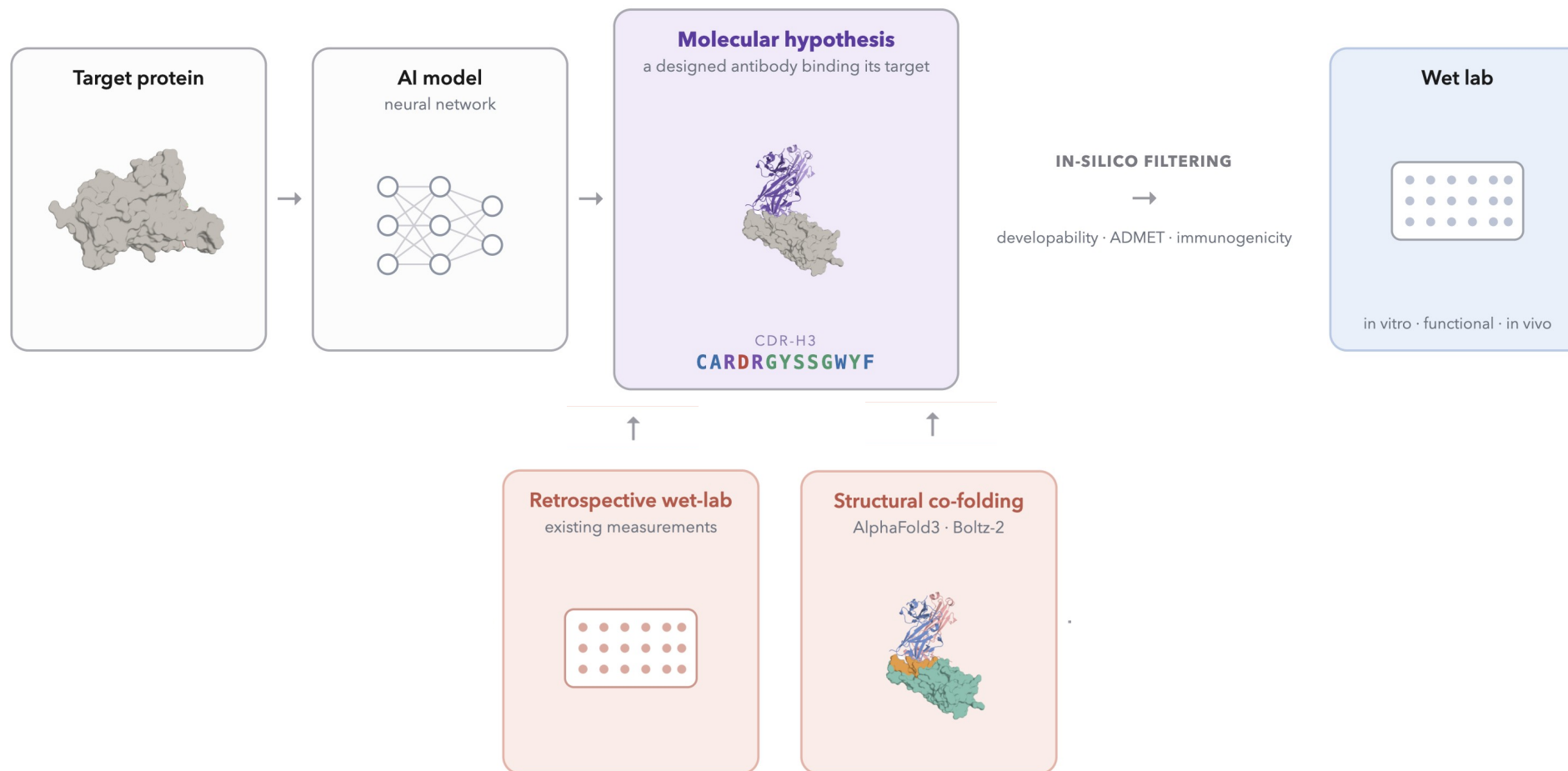
42 mio parameters

Encoder-decoder trained in 15 hours on a single GPU

Approximate cost of training < \$150

Publicly available (open-source) data

AI Drug Discovery is a Molecular Hypothesis





Derek Mason

Project Director / Director of Research Operations



Alexander Yermanos

Project Director Immunoglobulin Initiative / Assistant Professor, UMC Utrecht



Dr. Ulrike Haessler
Dr. Eddie Irvine
Dr. Ahmed Sadek
Dr. Edo Kapetanovic
Danielle Shlesinger
Dzhansu Hasanova
Elisa Marani
Fabrice Schlatter
Jami Han
Joshua Kämpf

Kai-Lin Hong
Kai-Ling Wang
Kevin Letscher
Luca Marchetti
Marco Mandolesi
NingNing Chen
Rachita Kumar
Riyaz Khan
Rok Krese
Lucas Stalder
Stefan Homentcovschi
Vittoria Martinolli
Wiona Glänzer

Computational Convergence of Adaptive Immunity and Artificial Intelligence

Sai T. Reddy

Posted February 05, 2026.

Methods for Molecular Recognition Computing

Sai T. Reddy

Posted April 03, 2026.

Contrastive learning for antibody-antigen sequence-to-specificity prediction

Hyunho Lee, Karla Castro, Sean Renwick, Lucas Stalder, Wiona Glänzer, Rachita Kumar, NingNing Chen, Andreas Scheck, Alex Yermanos, Derek Mason, Sai T. Reddy

Posted February 26, 2026.

Vibe Coding Specificity Foundation Models

Sai T. Reddy

Posted June 04, 2026.

A Drug-Target Specificity Foundation Model for Off-target Prediction, Repurposing, and Generative Design

Sai T. Reddy

Posted June 08, 2026.

Generative Drug Design in a Loop with dtSFM

Sai T. Reddy

Posted June 11, 2026.

All SFMs are Open-Source

Pre-trained model weights for all six SFMs are available on Hugging Face:

<https://huggingface.co/SFM-BIIE-ETHZ>.

Code, training and evaluation, pipelines, and orthogonal audit records are available on Github:

<https://github.com/Reddy-BIIE-ETHZ/Vibe-Coding-SFMs>.

Sequences of all designed variants are disclosed.

PDF + Vibe Coding Prompt, can run the model on an AI coding LLM (Claude Code, OpenAI Codex)

abSFM will be released later this summer after wet-lab validation experiments are completed

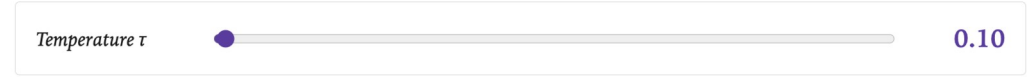
$$P(\text{bind } j) = \frac{\exp(s_j/\tau)}{\sum_k \exp(s_k/\tau)}$$

An antibody faces four candidate antigens. Each has a fixed compatibility score s_j (a dot product of CDR features and epitope features). Vary the temperature τ to see how the same scores become a different probability distribution.



Cold ($\tau = 0.1$) **Physiological ($\tau = 1.0$)** Hot ($\tau = 10$)

CANDIDATE	s_j	PROBABILITY	P_j
HEL <i>cognate target — hen egg lysozyme</i>	3.00		0.676
HEL mutant <i>1 CDR-contact residue swap</i>	2.00		0.249
BSA <i>off-target serum protein</i>	0.50		0.055
GFP <i>unrelated control</i>	-0.50		0.020



Cold ($\tau = 0.1$) Physiological ($\tau = 1.0$) Hot ($\tau = 10$)

CANDIDATE	s_j	PROBABILITY	P_j
HEL <i>cognate target — hen egg lysozyme</i>	3.00		1.000
HEL mutant <i>1 CDR-contact residue swap</i>	2.00		0.000
BSA <i>off-target serum protein</i>	0.50		0.000
GFP <i>unrelated control</i>	-0.50		0.000

The Binding Free Energy Landscape

$$P(j \mid i) = \frac{\exp(-E_{ij}/k_B T)}{\sum_{j'} \exp(-E_{ij'}/k_B T)}$$

$$E_{ij} = \Delta G = \Delta H - T \cdot \Delta S$$

